

Choice principles after the OpenAI partition-principle model

Complete research results and verification record

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Abstract

This report consolidates the results obtained in the present investigation of weak choice and cardinal ordering. Starting from OpenAI’s symmetric model of the partition principle without choice, it records a finite coinital cardinal basis theorem, cardinal well-foundedness and finite antichains, a native order-extension obstruction to the Boolean Prime Ideal Theorem, explicit compactness counterexamples, exact cardinal arithmetic and an all-base exponent-collapse phenomenon. It also presents audited parameter variants, Ramsey spectra, and a different extension obtained by forcing a linear order on the native partial order. The latter preserves substantial choice and covering properties, but its partition principle and global cardinal classification remain unresolved. Earlier permutation-model criteria, positive local constructions and counterexamples are retained with their precise scopes. Proofs and proof sketches are paired with a public supporting record. The new arguments have AI-assisted mathematical checks; they have not received independent human peer review. The report separates established source theorems, deductions, preliminary extensions and open questions, without claiming global priority.

Attribution. The construction solving the original PP-versus-AC problem belongs to OpenAI [8]. The present report concerns additional consequences and model investigations. Substantial OpenAI Codex assistance was used for proof development, checking, literature searches and writing. Human authorship does not imply that independent human verification has already occurred.

Scope of “complete.” This is a consolidated account of the mathematical results saved through the reporting cut-off, including negative results and unresolved obligations. It does not assert completion of the research programme or of the paused five-hour investigation. Longer derivations and independent AI audit records are distributed in the accompanying source archive. Earlier exploratory statements are superseded by the statuses in this report. The original imported conversation is preserved privately, unchanged; it is not reissued as a new theorem paper.

Contents

1	Reading the results and keeping the models distinct	4
1.1	Notation and status conventions	4
1.2	Three settings	4
1.3	Main outcomes at a glance	5
1.4	Historical motivation	5
2	Finite bases, NDS and cardinal antichains	6
2.1	Extracting the cardinal normal form	7
2.2	A finite basis and its consequences	7
3	Cardinal order, exponentiation, and choice in the original model	9
3.1	The generator and the exact cardinal records	9
3.2	Lattice structure, finite bases, and finite powers	10
3.3	Supported names and exact exponentiation	11
3.4	Complete arithmetic table over the constructible ground	12
3.5	Choice consequences and their exact boundaries	13
3.6	Ramsey principles: ordinary truth and stronger separations	14
3.7	Comparability, cancellation, and historical boundaries	15
4	Boolean prime ideals, compactness, and model variations	17
4.1	A partial order with no linear extension	17
4.2	Explicit Boolean-algebra and graph witnesses	18
4.3	BPI, small violations of choice, and limits of compactness	19
4.4	Parameter changes and preservation	20
4.5	Exact pair and finite-arity Ramsey spectra	21
4.6	No-go results for repairing the model	22
4.7	Source corrections, attribution, and remaining boundaries	23
5	The ordinary extension by finite linear orders	24
5.1	Construction and the chain conditions	24
5.2	Preservation and the remaining failures of Choice	25
5.3	The new real set and the unresolved evaluation quotient	26
5.4	Local real fields and exact localization	27
5.5	All comparisons between old sets are preserved	27
5.6	Explicit nonsplitting and additional choice bounds	28
5.7	Initial segments, final segments, and open intervals	28
5.8	Boundaries of the current classification	29
6	Earlier permutation-model results and construction obstructions	31
6.1	Exact and sufficient group criteria	31
6.2	Necessary topology and a sharp atom-model boundary	32
6.3	Germ and monoid completions: what succeeded locally	32

6.4	Explicit counterexamples to the germ candidates	33
6.5	Fields, local-motion groups, and symmetric iterations	34
6.6	Coherent coding is possible, but the constructed models fail	35
6.7	Other imported forcing results and their limits	35
7	Verification, novelty and the remaining research programme	36
7.1	What was checked formally	36
7.2	Mathematical and finite checks	36
7.3	Novelty boundaries	37
7.4	Open questions retained at the cut-off	37
7.5	Files, authorship and publication record	38

1 Reading the results and keeping the models distinct

1.1 Notation and status conventions

All unqualified mathematical implications are over classical ZF. We write $X \preccurlyeq Y$ for an injection $X \hookrightarrow Y$, $X \cong Y$ for a bijection, and $X \prec Y$ for $X \preccurlyeq Y$ with no reverse injection. These comparisons concern cardinalities; they do not require a globally chosen system of cardinal representatives. The Hartogs number $h(X)$ is the least ordinal not injecting into X . The Lindenbaum number $\ell(X)$ is the least nonzero ordinal not a surjective image of X , with harmless empty-set conventions handled separately. Under AC_{WO} , the largest well-ordered cardinal injecting into an infinite set X exists; it is denoted $w(X)$.

Source theorem.

A result in a cited external manuscript or published work. OpenAI's construction remains an explicit dependency.

Audited deduction.

An argument reconstructed or developed in this project and checked mathematically by separate AI agents and/or the coordinating agent. This is not a claim of external peer review or full formalization.

Conditional result.

A proved implication with additional stated hypotheses whose realization or preservation may remain open.

Unresolved here.

No proof or counterexample has been obtained in this investigation. This label does not assert that the world literature treats the problem as open.

The partition principle is

$$\text{PP} : \quad X \twoheadrightarrow Y \implies Y \hookrightarrow X.$$

The injection need not be a section of the given surjection. Every surjection having a section is equivalent to AC , and must not be substituted for PP . Choice for all ordinal-indexed families is AC_{WO} ; classical implications give $\text{PP} \implies \text{AC}_{\text{WO}} \implies \text{DC}$. Choice for arbitrary families of finite or countable *members* is a different assertion from choice for finite or countable *index sets*.

BPI denotes the Boolean Prime Ideal Theorem, equivalently the existence of an ultrafilter on every nontrivial Boolean algebra. OP says every set can be linearly ordered. OEP says every partial order has a linear extension. The implication $\text{BPI} \implies \text{OEP} \implies \text{OP}$ is classical; the two order principles must not be identified. The ordinary Ramseyan partition principle is denoted RPP and is defined in the cardinal chapter; it is not the Ramsey property of a group or normal filter.

1.2 Three settings

1. **The original pure model W .** This is the symmetric model in the pinned OpenAI source. The ground V satisfies ZFC . The distinguished non-well-orderable set A consists of Cohen subsets of a regular ordinal κ . In the original construction $\kappa = \omega_1$ and the ground parameter is $\mu = ((2^{\aleph_0})^+)^V$. The monoid T has ground cardinality μ . For this original model, the audited internal identification is $\mu = \omega_2^W$.
2. **The finite-order extension $N = W[G]$.** Here G is generic for finite linear orders on subsets of A extending its native partial order. The generic order exists in N , while it

provably does not exist in W . Thus the original order-extension obstruction is removed. This does not establish global BPI or preserve the original image classification.

3. **Ordinary permutation models with atoms.** Earlier work used a faithful group G and an effective normal filter of subgroups to build ZFA models. Its group criteria and counterexamples concern that setting. A pure symmetric extension is not an ordinary atom permutation model; the relevant collapse and transfer arguments cannot be silently moved between them.

The source revision throughout is

adc7f1241b42e322a6451854ab7e4b4c146bf78a.

The source supplies $\text{SVC}(A)$, AC_{WO} , A non-well-orderable, and the property

$Q(A)$: every non-well-orderable surjective image of A is bijective with A .

The anchors are Proposition 3.6, Proposition 3.7, Proposition 3.8, Theorem 6.3 and Lemma 7.1 of [8]. Section 8 supplies the ordinary first-order relative-consistency bridge. None of our applications replaces this bridge by an assumption that a transitive model or an inaccessible cardinal exists.

1.3 Main outcomes at a glance

Setting	Recorded outcome
Original W	$\text{PP} + \text{AC}_{\text{WO}} + \text{DC} + \text{FB} + \text{NDS} + \neg\text{AC}$; all set-sized cardinal antichains finite, but their finite sizes unbounded.
Original W	KWP_1 , OP , choice for finite members, and choice for arbitrary families of countable members; $\neg\text{DC}_{\omega_1}$.
Original W	Native partial order has no linear extension; consequently $\neg\text{OEP}$ and $\neg\text{BPI}$. Explicit Boolean and graph witnesses have size A .
Original W	Exact cardinal normal forms and coordinate comparison; set-sized suprema and nonempty infima; all infinite X satisfy $X^3 \cong X^2$.
Original W	For every non-well-orderable X and every set Y , $Y^X \cong Y^{w(X)}$. In particular $\mu \prec A$ while $Y^\mu \cong Y^A$ for every Y .
Variants	Regular-parameter adaptations separate ordinal-comparability and Ramsey thresholds while retaining the strong original package and failure of BPI.
Extension N	$\text{AC}_{\text{WO}} + \text{DC} + \neg\text{AC} + \neg\text{DC}_{\omega_1}$; old cardinals and cofinalities preserved; $\text{SVC}(A)$ and $\text{SVC}(\mathbb{R})$; a non-well-orderable continuum.
Extension N	$A \hookrightarrow \mathbb{R}$, $A \twoheadrightarrow \mathbb{R}$, and $A^\omega \cong \mathbb{R}$; the two displayed directions do not give $\mathbb{R} \hookrightarrow A$. Every generic open interval and final segment has size A .
Unresolved	General $\text{PP} \Rightarrow \text{NDS}$; a different model of $\text{PP} + \text{BPI} + \neg\text{AC}$; $\text{PP}, Q(A), \text{FB}, \text{NDS}, \text{BPI}$ in N ; and $\mathbb{R}^N \hookrightarrow A$.

1.4 Historical motivation

The partition principle goes back to the early development of choice principles; a historical account and modern formulation appear in [4, 1]. Howard and Tachtsis studied no decreasing sequence of cardinals [6], and Tachtsis studied its position in the weak-choice hierarchy [12].

Blass and Kulshreshtha distinguish 24 well-foundedness formulations and identify them under PP [3]; that equivalence does not prove any of them from PP alone.

Feldman–Orhon, with Blass’s appendix, show that any fixed finite bound on cardinal antichain size forces choice [5]. The assertion that each antichain is finite has no fixed bound and is different. Karagila explicitly records the corresponding question [7]; Tachtsis studies related infinite-family comparability [11]. Our use of the label FB is descriptive. The finite-product order argument is classical well-quasi-order theory [14, 16]; the application to the cited construction is the additional deduction.

2 Finite bases, NDS and cardinal antichains

All arguments take place in classical ZF. Write $X \preceq Y$ if there is an injection $X \hookrightarrow Y$, and $X \prec Y$ if $X \preceq Y$ and $Y \not\preceq X$. Write $X \cong Y$ for the existence of a bijection. The disjoint union is denoted by $\sqcup$.

Definition 2.1. The following are statements about sets.

- (i) NDS: there is no sequence $(X_n)_{n < \omega}$ with $X_{n+1} \prec X_n$ for every n .
- (ii) The finite-antichain principle: every set family $\mathcal{F}$ whose distinct members have incomparable cardinalities is finite.
- (iii) FB: for every nonempty set family $\mathcal{F}$, there is a nonempty finite $\mathcal{F}_0 \subseteq \mathcal{F}$ such that

$$\forall X \in \mathcal{F} \exists Y \in \mathcal{F}_0 \quad Y \preceq X. \quad (1)$$

No simultaneous choice of all the injections in (1) is part of the assertion. Likewise, NDS quantifies over sequences of sets, not sequences already equipped with injection witnesses. This is the formulation $WF_{i_3}^i$ of [3]; distinct formulations must not be interchanged in bare ZF.

For statements phrased directly about cardinalities, one may use Scott representatives: the set of all representatives of the least possible rank. If $\mathcal{C}$ is a set of Scott cardinalities, its union is a set of representatives. Applying FB to that union avoids choosing one set for each cardinality.

Definition 2.2. Fix a nonempty set A . The *homogeneous-image hypotheses* are:

- (A) SVC(A): every nonempty set X is the image of a surjection $A \times \eta \twoheadrightarrow X$ for some ordinal η .
- (B) AC_{WO}: every ordinal-indexed family of nonempty sets has a choice function.
- (C) Every non-well-orderable surjective image of A is bijective with A .

Non-well-orderability of A itself is required for the consistency application, but not for the abstract finite-basis implication.

Theorem 2.3 (Central implication). *In ZF, the hypotheses of Definition 2.2 imply FB. Hence every sequence of sets has a forward-comparable pair, NDS holds, and every set-sized cardinal antichain is finite.*

The proof is given in Subsections 2.1–2.2. The resulting application is

$$\text{Con}(\text{ZF}) \implies \text{Con}(\text{ZF} + \text{PP} + \text{AC}_{\text{WO}} + \text{FB} + \text{NDS} + \neg \text{AC}), \quad (2)$$

with finiteness of all set-sized cardinal antichains in the same model. The construction in [8] is the external input for (2).

2.1 Extracting the cardinal normal form

Lemma 2.4. *Under Definition 2.2, every set X has a representation*

$$X \cong (A \times \alpha) \sqcup \beta, \quad \alpha, \beta \in \text{Ord}. \quad (3)$$

The two parameters may be chosen to be initial ordinals; their particular values need not be unique.

Proof. The empty set uses $\alpha = \beta = 0$. Otherwise fix one surjection $s : A \times \eta \rightarrow X$. Put

$$B_i = s[A \times \{i\}], \quad T_i = B_i \setminus \bigcup_{j < i} B_j \quad (i < \eta).$$

The T_i partition X . Each nonempty T_i is a surjective image of A : fix one $t \in T_i$, keep $s(a, i)$ when it belongs to T_i , and send all other arguments to t . This proves existence for one slice at a time; it does not choose default elements simultaneously.

Let I consist of those $i < \eta$ for which T_i is not well-orderable. For $i \in I$, hypothesis (C) supplies a bijection $A \rightarrow T_i$; for $i \notin I$, take a well-order of T_i , including the empty relation when the slice is empty. The candidate bijections lie in $\mathcal{P}(A \times X)$ and the candidate orders in $\mathcal{P}(X \times X)$. Thus AC_{WO} selects witnesses from a set-sized, well-orderably indexed family.

The union of the non-well-orderable slices is bijective with $A \times I$, hence with $A \times \alpha$ for an ordinal α . The remaining slices have a well-orderable union: order by slice index first, then by the selected well-order inside each slice. That union is bijective with an ordinal β . The two pieces are disjoint and cover X . Replacing each ordinal by its initial ordinal preserves the representation. $\square$

The slicing mechanism is an extraction of the method already used in [8, Lemma 7.1], which in turn credits Ryan-Smith's local-to-global argument [10]. The next finite-basis step is the additional application needed here.

For each X , choose its parameters *definably*: let $\alpha(X)$ be the least ordinal admitting any representation (3), and let $\beta(X)$ be the least ordinal witnessing it for that $\alpha(X)$. These exist because any one witness bounds a nonempty set in which to minimize. Replacement collects the pairs for any set family. No class choice or simultaneous selection of normal-form bijections is being made. Arbitrary representations need not be unique; only the chosen least pair is canonical.

2.2 A finite basis and its consequences

Lemma 2.5 (Ordinal pairs, in ZF). *Every nonempty set $R \subseteq \text{Ord}^2$ has a nonempty finite subset R_0 such that every pair in R is coordinatewise above a pair in R_0 .*

Proof. Choose the lexicographically least residual pair, delete its coordinatewise upper cone, and repeat while a pair remains. The rule is unique, so recursion on ω requires no choice. If it never stopped, the selected first coordinates would be nondecreasing: removal cannot introduce a smaller first coordinate. Since every later pair survived deletion of each earlier upper cone, its second coordinate would be strictly smaller than the preceding second coordinate. This would be an infinite descending sequence of ordinals. The process therefore stops after finitely many selections. Every deleted pair lies above a selected pair, proving the assertion. $\square$

Proof of the finite-basis assertion in Theorem 2.3. For a nonempty set family $\mathcal{F}$, apply Lemma 2.5 to

$$R = \{(\alpha(X), \beta(X)) : X \in \mathcal{F}\}.$$

Choose one member of $\mathcal{F}$ for each of the finitely many basis pairs. This is finite choice, a theorem of ZF. Let $\mathcal{F}_0$ be the resulting finite family. If the profile of $Y \in \mathcal{F}_0$ is coordinatewise below that of $X \in \mathcal{F}$, the coordinate inclusions give

$$(A \times \alpha(Y)) \sqcup \beta(Y) \hookrightarrow (A \times \alpha(X)) \sqcup \beta(X).$$

Composing with two normal-form bijections gives $Y \preceq X$. This proves (1). The injections need only exist separately. $\square$

Corollary 2.6. *Under FB, every sequence $(X_n)_{n < \omega}$ has indices $i < j$ such that $X_i \preceq X_j$. In particular, NDS holds.*

Proof. Apply FB to the range of the sequence. Each member of its finite basis has a least occurrence index. Choose j larger than all these indices. A basis member injects into X_j , and its least index i is smaller than j .

If the sequence were strictly decreasing, finitely many downward injections would give $X_j \preceq X_{i+1}$. Composing contradicts $X_i \not\preceq X_{i+1}$. Only finitely many witnesses are used, not countably many chosen injections. $\square$

Corollary 2.7. *Under FB, every set-sized cardinal antichain is finite. More generally, every nonempty set family represents finitely many minimal cardinalities, and each of its cardinalities lies above one of them.*

Proof. For an antichain, a basis member can inject into another member only when they represent the same cardinality. Hence the finite basis accounts for every cardinality in the antichain. For a general family, remove redundant cardinalities from a finite basis, leaving its minimal ones. They remain coinital. If a family cardinality lay strictly below one of them, another basis member would lie below that smaller cardinality, contradicting minimality in the finite basis. Conversely every minimal family cardinality must coincide with a basis cardinality. $\square$

There can still be infinitely many distinct sets representing a single minimal cardinality. Corollary 2.7 concerns the number of cardinalities, not the number of such representatives. The Scott-set formulation follows by applying FB to the union of the sets of representatives.

Relationship with well-foundedness. Over ZF, FB is equivalent to well-foundedness of the injection order on each set of cardinalities together with finiteness of all cardinal antichains. For the converse, the minimal members of a nonempty family form a finite antichain, and well-foundedness supplies a minimal member below each given member. Under DC, absence of an infinite strict descending sequence is equivalent to this set-sized well-foundedness: a nonempty downward set with no minimal member would yield a descending sequence by dependent choice. Consequently, over ZF + DC, FB is equivalent to NDS plus finite cardinal antichains. The qualification by DC cannot simply be erased.

Relative consistency. Applying the theorem to the stronger source package and retaining the source's non-well-orderable A yields $\text{Con}(\text{ZF}) \Rightarrow \text{Con}(\text{ZF} + \text{PP} + \text{FB} + \text{NDS} + \neg\text{AC})$. Under the same PP, the well-foundedness equivalence of [3, Proposition 9.8] gives the other 23 formulations. These are consequences in this model; neither $\text{PP} \Rightarrow \text{NDS}$ nor $\text{NDS} \Rightarrow \text{PP}$ is established generally here.

3 Cardinal order, exponentiation, and choice in the original model

This section collects the results established for the *original* symmetric model W , and, where explicitly indicated, its separately audited parameter variants. It does not transfer these conclusions to the later ordinary forcing extension. The foundational construction, its Partition Principle and support lemmas are due to OpenAI [8]; the additional calculations below are the research deductions reported here. They have received separate mathematical checks by AI agents, not external human peer review or a complete Lean formalization. Historical qualifications are given at the end of the section.

3.1 The generator and the exact cardinal records

Write $X \leq Y$ for an injection and $X \cong Y$ for a bijection. In the source's general ground $V \models \text{ZFC}$, put

$$\kappa = \omega_1^V, \quad \mu = ((2^{\aleph_0})^+)^V.$$

The set $A \subseteq \mathcal{P}(\kappa)$ consists of the distinct Cohen rows. The source proves $\text{SVC}(A)$, choice for ordinal-indexed families AC_{WO} , and the classification

$$\text{every non-well-orderable image of } A \text{ is bijective with } A. \quad (\text{Q})$$

Together these give PP and $\neg \text{AC}$ in W . The same source locality implies that a function on an ordinal with values in a ground set belongs to the Cohen intermediate extension. Thus ground cardinals at least μ remain cardinals in W .

The action and absorption. The generic diagrams define an internal free right action of the ground monoid T on A . For each a , the map $t \mapsto a \cdot t$ is injective; its image aT is a *chart*, of cardinal μ . Every ordinal-indexed subset of A lies in one chart. The latter assertion follows by copying a putative outside value freshly while fixing the supporting diagram: two compatible conditions would assign different labels to the same ordinal index. Hence the largest well-orderable cardinal injecting into A is

$$w(A) = \mu, \quad h(A) = \mu^+.$$

Here h is the Hartogs number. No global inverse-label map $A \rightarrow I$ is asserted to belong to W .

Each map $a \mapsto a \cdot t$ is onto. Define a predecessor diagram by $q(tx) = p(x)$ and put its remaining coordinates freshly; exact amalgamation makes such predecessors dense. For $t \neq 1$, every fiber over an anchor a_0 is non-well-orderable. To see the key step, normalize an alleged ordinal enumeration to a condition (d, p) containing its support and the Cohen support, with $a_0 = A_{p(ce_0)}$, where $e_0 \neq 1$ is the distinguished monoid element. Take $q_{e_0} = p$. The source probe clause at $(e_0c, t, \{1\})$ supplies n with

$$nt = e_0ce_0, \quad e_0 \not\sqsubseteq n,$$

where $u \sqsubseteq v$ means $uz = v$ for some z . Thus $A_{q(n)}$ is in the fiber but outside the root diagram. Decide its enumeration index and copy both the stronger diagram and its Cohen condition freshly over the root. Exact diagram overlap and disjoint Cohen supports off the root give the contradiction.

Fibers for distinct t are disjoint by freeness in the monoid coordinate. Each nonempty fiber is an image of A by a retraction, so (Q) identifies it with A . Choosing these bijections over the well-ordered set $T \setminus \{1\}$ gives $A \times \mu \leq A$. Consequently

$$A \times \mu \cong A, \quad A^2 \cong A. \quad (1)$$

For the second equality, first use the raw normal form below for A^2 . Its well-orderable subsets have well-orderable projections of size at most μ , so both ordinal parameters are at most μ and are absorbed by $A \times \mu$. This avoids assuming square absorption in advance.

Normal form and order. An $\text{SVC}(A)$ presentation $A \times \eta \rightarrow X$ splits X by its least presenting ordinal coordinate. By (Q), each resulting non-well-orderable piece is a copy of A ; the other pieces are well-orderable. Ordinal-indexed choice assembles the bijections, yielding $X \cong (A \times \alpha) \sqcup \beta$. Using (1), the exact records are

$$\begin{aligned} (0, \lambda) : & \quad \text{a well-orderable cardinal } \lambda, \\ (a, e), \mu \leq a \leq e : & \quad \begin{aligned} & A \times a \quad \text{if } a = e, \\ & (A \times a) \sqcup e \quad \text{if } a < e. \end{aligned} \end{aligned} \quad (2)$$

All nonzero coordinates here are ordinal cardinals. For the second kind, $w(X) = e$, $h(X) = e^+$, and

$$A \leq X, \quad e \leq X \leq A \times e.$$

Under AC_{WO} , the Lindenbaum number agrees with the Hartogs number. The least non-well-orderable cardinal is therefore $|A|$.

The records determine the *entire injection order*:

$$(a, e) \leq (c, f) \iff a \leq c \text{ and } e \leq f, \quad (3)$$

including the well-orderable records with first coordinate zero. For necessity in the first coordinate, an injection between non-WO normal forms cannot send an entire source A -slice into the target's ordinal tail. Ordinal-indexed choice picks one point from each source slice whose image lies in the target's A -part. These are distinct, so the source coefficient is at most the largest WO cardinal of that part, namely the target coefficient. Necessity for w is immediate. Conversely embed the A -part and the ordinal part using the two coordinate inequalities, then combine the embeddings using doubling of the target. The WO cases reduce to the definition of w .

3.2 Lattice structure, finite bases, and finite powers

Equation (3) gives a precise cardinal lattice, with necessary size qualifications: *every set-sized family* of cardinalities has a supremum, and every *nonempty set-sized family* has an infimum. They are obtained by coordinatewise suprema and minima in (2). The resulting pairs are again realizable records. Binary distributive laws follow from those for minimum and maximum in two ordinal chains. There is no largest cardinal and no empty infimum; this is not an unqualified claim of a complete lattice or of complete distributivity. Scott cardinals can be used throughout to avoid choosing representatives.

The finite-basis theorem proved elsewhere in this report says that every nonempty family has a finite cointial family under injections. It yields NDS and the absence of infinite antichains of cardinalities. It does *not* bound all antichains by one fixed natural number. For any positive n , the n records

$$(\mu^{+i}, \mu^{+(2n-i)}), \quad i < n,$$

form an antichain: the first coordinates increase and the second decrease. In bare ZF the exact order-theoretic equivalence is $\text{FB} \leftrightarrow \text{WF}_{\text{card}} + \text{FAC}$, where WF_{card} is minimal-element well-foundedness on sets of Scott cardinals and FAC means every such antichain is finite. With DC it becomes $\text{FB} \leftrightarrow \text{NDS} + \text{FAC}$. The representative and sequence distinctions should not be suppressed when DC is absent. Moreover, the already proved PP and NDS give all 24 cardinal well-foundedness variants in Blass–Kulshreshtha's Definition 4.1, using their equivalence under PP [3, Proposition 9.8]. This equivalence does not prove NDS from PP alone.

For a fixed ordinal cardinal λ , the assertion W_λ that every cardinal is comparable with λ has the exact threshold

$$W \models W_\lambda \iff \lambda \leq \mu. \quad (4)$$

Every non-WO set contains A , proving the positive direction; A is incomparable with every ordinal cardinal above μ , proving the negative. The usual surjective version agrees under PP. In particular full comparability still fails, as it must when AC fails.

Every infinite set absorbs a countable multiplier and doubles. For non-WO X with record (a, e) , direct multiplication gives

$$X^n \cong A \times e \quad (2 \leq n < \omega). \quad (5)$$

Indeed $A \times e \leq X^2 \leq A^2 \times e^2$, and multiplying the normal form by $A \times e$ leaves the same cardinal. Thus $X^3 \cong X^2$ holds for every infinite X , but $X^2 \cong X$ does not: $X = A \sqcup \mu^+$ has record (μ, μ^+) and its square has record (μ^+, μ^+) . The question about universal square/cube stability is historical; Rubin–Rubin’s 1985 introduction already discusses it. The bounded search does not establish its intervening publication status.

There is also a finite bound, depending on X , on iterated power-set roots: some $N < \omega$ satisfies

$$\mathcal{P}^n(Y) \cong X \implies n \leq N. \quad (6)$$

Represent every possible root by a subset of X , using iterated singleton maps. The resulting set of roots has a finite coinital basis $Y_1, \dots, Y_k$, with unique depths $n_1, \dots, n_k$; uniqueness follows from Cantor’s theorem. If a root Y had depth $m > \max n_i$, some $Y_i \leq Y$ would give $X \cong \mathcal{P}^{n_i}(Y_i) \leq \mathcal{P}^{n_i}(Y) < \mathcal{P}^m(Y) \cong X$, impossible. In particular every cardinal is a finite power-set iterate of one which is not a power-set cardinal. There is no bound uniform in X .

3.3 Supported names and exact exponentiation

For ground ordinals b, λ , write

$$E_V(b, \lambda) = |\{f \in V : f : \lambda \rightarrow b\}|^V.$$

This is a ground cardinal; unmarked exponentials are computed in W . The decisive local calculation, for $b \geq 2$ and ordinal cardinals $\lambda \geq \mu$, is

$$b^A \cong A \times E_V(b, \mu), \quad b^\lambda \cong A \times E_V(b, \lambda). \quad (7)$$

The coefficients are at least $(2^\mu)^V$ and are preserved cardinals.

Here are the ingredients ensuring that (7) counts *all internal* well-orderable families. Fix a supporting diagram p . Normalize names to globally total A -to- b functions, using the zero function whenever the original value has the wrong type. The canonical graph name uses only label/ordinal pair names and the ordinary forcing relation. It is equivariant, keeps the support, and has a fixed rank bound. Thus these names form a ground set, rather than a class of unbounded-rank names. For each $t \in T$ choose a ground diagram q^t with $(q^t)_t = p$. At the record (t, s, d) , with d a Cohen condition on D_{q^t} , record $\xi < b$ exactly when

$$(d, q^t) \Vdash \tau(A_{q^t(s)}) = \xi.$$

There are μ positions, and at most one value per position. The code is a *partial value function*, so there are at most $(|b|^V + 1)^\mu = E_V(b, \mu)$ codes in the ground.

Equal codes force equality below $(1, p)$. Otherwise put an input where the values differ into a larger diagram r , write $p = r_t$, restrict the Cohen witness to D_r by source locality, and transport it to q^t by a permutation fixing D_p . It produces different values at the same record, a contradiction.

Ground choice now selects one representative per code. Their ordinal enumeration is HS with support D_p . Any internal WO family of functions has one supported enumeration; normalizing its coordinates with the same support embeds the whole family into that representative list by least indices. Ground functions on one chart give the matching lower bound. For ordinal domain λ the records are (d, ξ) , $\xi < \lambda$, and both local-decision clauses give the analogous count. There is no maximum-principle assumption, choice indexed by A , or automorphism of the actual pure model in this argument. Finally both function sets are non-WO and square-idempotent; (2)–(5) turn their exact WO bounds into (7).

Universal exponent collapse. For every non-WO X , set $e = w(X)$. Then for *every* set Y ,

$$Y^X \cong Y^e. \quad (8)$$

More explicitly,

$$\begin{aligned} \mathcal{P}(X) &\cong A \times E_V(2, e) \cong \mathcal{P}(e), \\ b^X &\cong A \times E_V(b, e) & (b \geq 2 \text{ ordinal}), \\ Y^X &\cong A \times E_V(f, e) & (Y \text{ non-WO, } f = w(Y)). \end{aligned} \quad (9)$$

The same formulas hold with domain e .

For the proof, two-point coding and $A \leq \mathcal{P}(\kappa)$ give $A^e \cong \mathcal{P}(e)$. The normal form also gives $\kappa \times X \cong X$, hence $A^X \cong \mathcal{P}(X)$. Sandwich an ordinal-base power between b^e and $b^{A \times e}$. With $\delta = E_V(b, \mu)$, (7) calculates the upper endpoint as

$$(b^A)^e \cong (A \times \delta)^e \cong A^e \times \delta^e \cong A \times E_V(b, e),$$

since ground arithmetic has $E_V(\delta, e) = E_V(b, e)$. The lower endpoint has the same cardinal. Its case $b = 2$ proves the power-set formula before that formula is used for non-WO bases. For non-WO Y , use $A \leq Y$, $f \leq Y \leq A \times f$ for the upper bound. To inject $A \times (f^e)^V$ into Y^e , reserve one coordinate for the A point and e further coordinates for the ground function, using $e + 1 \cong e$. This provides the matching lower bound in both Y^e and Y^X . Empty and singleton bases give the trivial equalities because X and e are nonempty. This exhausts all internal bases.

Consequently

$$\mu < A, \quad \forall Y \ (Y^\mu \cong Y^A). \quad (10)$$

The statement asserts pointwise existence of bijections. It asserts neither a natural transformation between function-space functors nor one class-sized choice of all bijections.

3.4 Complete arithmetic table over the constructible ground

For the separately audited L -ground variants $M_{m,n}$, put $\kappa = \aleph_m$, $\mu = \aleph_n$, where $1 \leq m < n < \omega$. All alephs are preserved. The original construction over L is $M_{1,2}$. The following tables concern these original symmetric models. The records of power sets are

$ X $	$ \mathcal{P}(X) $ as a record
finite r	$(0, 2^r)$
infinite WO $\lambda < \kappa$	$(0, \lambda^+)$
WO $\kappa \leq \lambda < \mu$	(μ, μ)
WO $\lambda \geq \mu$	(λ^+, λ^+)
non-WO (a, e)	(e^+, e^+)

(11)

For the small ordinal plateau, the local Cohen forcing has size μ and the κ^+ chain condition. Nice names for subsets of λ number at most $\mu^{\max(\kappa, \lambda)}$, equal under ground GCH to μ for $\kappa \leq \lambda < \mu$

and to λ^+ for $\lambda \geq \mu$. One supported enumeration bounds every WO family; a chart and the ground power set supply the lower bounds. Below κ there are no new subsets of ordinals. In particular, in $M_{1,2}$,

$$\mathbb{R} \cong \aleph_1, \quad \mathcal{P}(\aleph_1) \cong A, \quad \mathcal{P}(A) \cong \mathcal{P}(\aleph_2) \cong A \times \aleph_3.$$

This is a ground-GCH calculation, not GCH inside the choiceless model.

For completeness write $E(b, \lambda) = E_L(b, \lambda)$. Apart from empty and singleton cases, the *entire exponent table* is

$ Y $	$ X $	$ Y^X $ as a record
WO $b \geq 2$	WO $2 \leq \lambda < \kappa$	$(0, E(b, \lambda))$
WO $b \geq 2$	WO $\lambda \geq \kappa$	(θ, θ)
non-WO (a, f)	WO $\lambda \geq 2$	$(E(f, \lambda), E(f, \lambda))$
WO $b \geq 2$	non-WO (c, e)	$(E(b, e), E(b, e))$
non-WO (a, f)	non-WO (c, e)	$(E(f, e), E(f, e))$

(12)

where $\theta = \max(\mu, E(b, \lambda))$. For $X = \emptyset$ the result is 1, including $0^0 = 1$; for $X \neq \emptyset$ and $Y = \emptyset$ it is 0; a singleton base always gives 1; a singleton exponent returns Y with its original record. This last exception matters when $a < f$.

To justify the rows not already covered by (9), ordinal-valued functions of length $\lambda < \kappa$ are ground functions. Also $A^\lambda \cong A$ for $0 < \lambda < \kappa$: a sequence has range in one chart and its T -coordinate sequence is ground, giving $A \times (T^\lambda)^V \rightarrow A^\lambda$; now $\mu^{<\kappa} = \mu$, (1), and PP apply. For $\lambda \geq \kappa$, nice function names number at most $(\max(\mu, b)^\lambda)^L = \max(\mu, E(b, \lambda))$. This gives the WO-base row as above. The non-WO-base upper bound is $A^\lambda \times f^\lambda$. A matching lower bound uses two coordinates if λ is finite, since $f^\lambda = f$ in the ground, and the reserved-coordinate coding if λ is infinite.

The ground coefficients in (12) are explicit. For positive finite λ , $E(b, \lambda)$ is the usual finite power when b is finite and is b when b is infinite. For infinite λ and $b \geq 2$,

$$E(b, \lambda) = \begin{cases} \lambda^+, & b \leq \lambda, \\ b, & \lambda < b \text{ and } \lambda < \text{cf}^L(b), \\ b^+, & \text{cf}^L(b) \leq \lambda < b. \end{cases} \quad (13)$$

The final case concerns singular b ; it follows from König's theorem and $b^\lambda \leq 2^b = b^+$.

3.5 Choice consequences and their exact boundaries

In addition to PP, AC_{WO} , FB and NDS, the original model satisfies ordinary DC, KWP_1 , the Ordering Principle, and choice for arbitrary families of nonempty finite sets. The classical implication $\text{AC}_{\text{WO}} \Rightarrow \text{DC}$ is Jensen's theorem as presented in Jech [15, Theorem 8.2]; an earlier apparent source conflict was resolved by reading the actual PDF. For KWP_1 , PP turns a presentation into $X \leq A \times \eta$, which embeds into a power set of an ordinal by tagged coding. Lexicographic order on that power set linearly orders X . Least elements of finite subsets then give finite-member choice. This does not assert that every prescribed partial order has a linear extension: the separate obstruction in this report proves failure of that stronger principle and of BPI.

The source itself explicitly states the dual Cantor–Bernstein principle CSB^* and Weak Partition Principle as consequences of PP. With $X \leq^* Y$ meaning $X = \emptyset$ or $Y \twoheadrightarrow X$, these say respectively

$$X \leq^* Y \ \& \ Y \leq^* X \Rightarrow X \cong Y, \quad X \leq^* Y \Rightarrow \neg(Y < X).$$

They are not newly extracted results of this report.

Choice for countable members. A further saved proof, now independently audited twice, gives

$$\mathcal{C}(\infty, \leq \aleph_0) : \text{ every family of nonempty countable sets has a choice function. } \quad (14)$$

This is Howard–Rubin Form 85A. The family of indices is arbitrary; its *members* are countable, so (14) differs from countable choice. For one nonempty countable $S \subseteq A \subseteq \mathcal{P}(\omega_1)$, its pairwise first-difference ordinals are countably many and therefore bounded. Let $\alpha(S) < \omega_1$ be the least ordinal whose restriction map separates S . Source preservation gives $\mathcal{P}(\alpha)^W = \mathcal{P}(\alpha)^V$ for every $\alpha < \omega_1$. One fixed ground family of well-orders of these power sets selects the least restriction and hence a unique member of S . No simultaneous choice of enumerations of S was used. Given arbitrary X , choose one embedding $X \leq A \times \eta$ and first take the least ordinal slice of a countable subset, then this selector on A . For an arbitrary family, apply the construction just once to its union. The proof is for original W ; new reals prevent an automatic transfer to the later extension.

Ordinary versus higher dependent choice. Despite DC, the original model and the audited parameter family fail DC_{ω_1} . Order A natively by $a \leq_T b$ iff $a \cdot t = b$ for some $t \in T$. Every ordinal-indexed subset has a common lower bound by chart locality, and every point has a strict predecessor by surjectivity of the e_0 -translation. There is no strictly decreasing ω_1 -sequence: localize its whole range to one chart, lift to T , and apply the source’s strictly increasing real-valued height. This would give a decreasing ω_1 -sequence of reals, impossible because its disjoint successive intervals inject into the rationals. Thus the tree of descending sequences of countable ordinal length is countably closed and has no terminal node but has no ω_1 -branch. Equivalently, the relation “strictly below every member of the preceding countable set” directly violates the conclusion of Howard–Rubin’s Form 44. Increasing only the Cohen parameter does not remove this height obstruction.

3.6 Ramsey principles: ordinary truth and stronger separations

Here RPP means the *Ramseyan Partition Principle*: every two-coloring of pairs of an infinite set has an infinite homogeneous subset. This is not PP and not the group-filter Ramsey condition used in FM analyses of BPI. Ordinary RPP holds in W : DC gives a countably infinite subset, and the ordinary finite-arity Ramsey theorem on ω is provable in ZF. More generally, NDS implies that every infinite set is Dedekind-infinite: otherwise finite injective lists from an infinite Dedekind-finite set produce a strictly descending sequence by excluding successively shorter lists. Thus both the PP/DC and NDS routes give RPP. These implications are classical weak-choice context [24, 3].

The stronger cardinal demands differ. Let NR_λ say that every two-coloring of pairs of every non-WO set has a homogeneous subset of cardinality λ . Throughout the audited parameter family,

$$\text{NR}_\lambda \iff \lambda \leq \kappa \quad (\lambda \text{ infinite WO}). \quad (15)$$

For the positive part pull a coloring back to a κ^+ -subset of A . An inner $L[C]$ containing its ordinal code and the required small sequence enumerations satisfies the appropriate Erdős–Rado theorem; its witness remains in W . The first-difference parity coloring on $A \subseteq \mathcal{P}(\kappa)$ gives the negative part: fresh-copy arguments make every homogeneous set WO, and a Cohen-condition Δ -system argument bounds its cardinal by κ . Each color also has a homogeneous set of size κ . The arbitrary-ground pair proof uses the separately audited collapse of $2^{<\kappa}$ to κ in W ; the L -ground cases need only their unchanged smaller power sets.

The color-count boundary is also exact. For every ordinal cardinal $1 \leq \eta < \kappa$, every η -coloring of pairs of a non-WO set has a κ -sized homogeneous subset. At $\eta = \kappa$ the coloring

$\{a, b\} \mapsto \min(a \triangle b)$ has no homogeneous triple: three distinct binary bits cannot be pairwise different at one coordinate. Thus κ is the least WO number of colors permitting such a failure.

For the L -ground family $M_{m,n}$, let $\text{NR}_\lambda^{r,\text{fin}}$ assert that every coloring of $[X]^r$ into any positive finite number of colors, for every non-WO X , has a homogeneous set of cardinality λ . Its exact spectrum is

$$M_{m,n} \models \text{NR}_\lambda^{r,\text{fin}} \iff \lambda \leq \aleph_{\max(m-r+2,0)}, \quad r \geq 2. \quad (16)$$

Here λ ranges over infinite WO cardinals. Inner-model Erdős–Rado supplies the positive bound; a generalized Sierpiński coloring and first-difference stepping up give the negative bound [25, 26]. The proof checks that homogeneous sets in the stepping-up construction are well-ordered by lexicographic order or its reverse, rather than assuming this in advance. The triple obstruction uses *four colors*. The exact two-color triple question remains unresolved here; the audit of simple binary coarsenings does not settle it.

For example, $M_{1,3}$ and $M_{2,3}$ have the same $w(A) = \aleph_3$, $h(A) = \aleph_4$, and the same entire threshold (4). Their pair homogeneous thresholds are $\aleph_1$ and $\aleph_2$; their finite-color triple thresholds are $\aleph_0$ and $\aleph_1$. Both retain PP, FB, NDS, AC_{WO} , DC and failure of DC_{ω_1} and AC. This separates the stronger Ramsey statements within that shared package. It is not a claim that the classical colorings or partition calculus are new.

3.7 Comparability, cancellation, and historical boundaries

All power sets in W have comparable cardinalities. An infinite set doubles, so its power set squares to itself. Every non-WO power set therefore has record (θ, θ) , and these form a chain. A WO power set must have a WO base of cardinal below κ ; otherwise it contains $A \leq \mathcal{P}(\kappa)$. Such smaller powers have ground enumerations below μ , so lie below every non-WO power. Thus

$$\text{PC} : \quad \forall X, Y [\mathcal{P}(X) \leq \mathcal{P}(Y) \text{ or } \mathcal{P}(Y) \leq \mathcal{P}(X)]. \quad (17)$$

Bare $\text{PC} + \neg \text{AC}$ is prior art: Glazer’s publicly dated 30 September 2025 answer proves it in Cohen’s first model [17]. The joint PP/FB/DC package and exact coefficients are the application under discussion here; no exhaustive novelty guarantee is claimed. In ZF, PC also implies KWP_1 : compare $\mathcal{P}(X)$ with $\mathcal{P}(\theta)$ for $\theta = h(\mathcal{P}(X))$. The latter cannot inject into the former, so $X \leq \mathcal{P}(X) \leq \mathcal{P}(\theta)$. This observation appears in Wojowu’s comment on the cited question.

The Injective Continuum Function assertion $\text{ICF} : \mathcal{P}(X) \cong \mathcal{P}(Y) \Rightarrow X \cong Y$ fails already in ZFC at a permissible continuum plateau, and is not a consequence of AC. Here it fails more specifically at the strict pair $\mu < A$, by (10). The order-reflecting variant $\mathcal{P}(X) \leq \mathcal{P}(Y) \Rightarrow X \leq Y$ fails as well. For finite Kinna–Wagner principles,

$$\text{ICF} + \text{KWP}_{n+1} \Rightarrow \text{KWP}_n \quad (n < \omega), \quad (18)$$

so ICF plus any finite KWP_n implies AC. Apply KWP_{n+1} to $\mathcal{P}(X)$ to get $\mathcal{P}(X) \leq \mathcal{P}(B)$ with $B = \mathcal{P}^n(\theta)$, θ infinite. Since B doubles, $\mathcal{P}(X \sqcup B) \cong \mathcal{P}(B)$; ICF gives $X \leq B$. Here $\text{KWP}_0 = \text{AC}$. No transfinite version or priority claim is asserted [23, 19]. The corresponding finite surjective principles KWP_n^* imply KWP_{n+1} by taking inverse images under the presentation surjection. Thus ICF combined with any finite KWP_n^* implies AC as well.

For clarity, distinguish two all-base assertions: (EC) if $Y^X \cong Y^Z$ for every Y , then $X \cong Z$; and (SES) if $X < Z$, some Y satisfies $Y^X < Y^Z$. These are descriptive labels, not asserted standard names. Both hold in ZFC. For infinite cardinals $\alpha < \beta$, choose $\nu = \beth_{\alpha^+}$, a strong limit of cofinality α^+ ; then $\nu^\alpha = \nu < \nu^\beta$ by bounded ranges and König’s theorem. Finite cases use base 2. Equation (10) refutes both in W .

Bare comparable all-base plateaux have an older mechanism. Mutual surjections induce injections between function spaces by precomposition at every base. From an infinite Dedekind-finite set D , let S be its finite injective lists. Deleting the first entry, with the empty list sent

to an extra point, maps S onto $S \sqcup 1$; a retraction maps back. But $S < S \sqcup 1$, so their powers agree at every base. Hamkins’s 2012 answer records this list construction; the all-base conclusion is the immediate precomposition extension [20]. Thus bare failure of EC or SES is not a new consistency phenomenon. The distinction in (10) is its compatibility with PP, DC and dual CSB, and its one well-orderable exponent; it cannot arise from unequal mutually surjective sets. For the logical record, ZF proves $\text{EC} \Rightarrow \text{SES}$: on a comparable nonempty pair, function extension gives the forward injections for every nonempty base; if none were strict, EC would give equality of the exponents. An empty smaller exponent is separated by base 2. It also proves $\text{EC} \Rightarrow \text{CSB}^*$, by the precomposition argument, and either EC or SES implies that every infinite set is Dedekind-infinite, by the finite-list counterexample just given. No converse or general equivalence with AC is established here. Finally strict *reflection*, $p^x < p^y$ and $p \neq 0 \Rightarrow x < y$, is a different statement, listed as equivalent to AC by Lindenbaum–Tarski [21, No. 82(A5)]. Its direction and quantifiers do not identify the strength of EC or SES.

The general implication $\text{PP} \Rightarrow \text{NDS}$ has not been proved by these model calculations. Nor do they prove PP alone implies FB, the exact arithmetic, or the stronger Ramsey statements. Earlier phase notes that left BPI in this concrete model undecided are superseded by the separate negative result in this report. Positive BPI in a different PP model, and the global choice classification of the later extension, remain distinct questions.

Proof-file map and verification scope. Detailed records are in `evidence/original_model/`: `T_ACTION_FIBERS_AND_SQUARE_COEFFICIENT_DRAFT.txt`; `T_ACTION_INDEPENDENT_AUDIT_AND_DC_OMEGA1.txt`; `EXACT_EXPONENTIALS_AND_UNIVERSAL_EXPONENT_COLLAPSE_AUDIT.txt`; `COMPLETE_GCH_GROUND_EXPONENT_TABLE.txt`; `FB_NDS_FAC_EXACT_CHOICE_AUDIT.txt`; `COUNTABLE_MEMBER_CHOICE_SECOND_INDEPENDENT_AUDIT.txt`; `GLOBAL_NON_WELL_ORDERABLE_RAMSEY_THRESHOLD.txt`; `HIGHER_ARITY_ROOT_INDEPENDENT_AUDIT.txt`; and `ALL_BASE_EXPONENT_CANCELLATION_LITERATURE_AUDIT.txt`. The lattice and root-depth proofs are in `evidence/finite_basis/PROOFS_cardinal_lattice_and_power_roots.txt`. These files preserve source dependencies and search limits. The new results in this section are mathematical deductions, not a certification of novelty or a claim that all of them have been formally verified.

4 Boolean prime ideals, compactness, and model variations

This section records consequences of the released OpenAI construction and adaptations checked in the present research. The construction itself is credited to OpenAI, at commit [adc7f124](#). The new arguments below have mathematical drafts and separate internal AI-agent audits. That is distinct from external peer review and from a formal proof of their full statements. Their novelty has not been certified. Older working notes saying that BPI in this model is “undecided” are superseded by the six-cycle theorem below.

Write $X \simeq Y$ for equipotence, and $X \preceq Y$ for the existence of an injection. The relevant established package in the original model W is

$$\text{ZF} + \text{PP} + \text{AC}_{\text{WO}} + \text{DC} + \text{FB} + \text{NDS} + \neg\text{AC}.$$

Here FB is the finite cardinal basis property defined earlier, not an abbreviation for finite choice. The additional structural inputs are $\text{SVC}(A)$, the quotient dichotomy $Q(A)$, and the least non-well-orderable cardinal $|A|$. Every set embeds into $A \times \eta$ for an ordinal η , every non-well-orderable set contains a copy of A , and $A^{<\omega} \simeq A^2 \simeq A$. The right action of the ground monoid T on A is free on each orbit. The native order is

$$a \leq_T b \iff (\exists t \in T) a \cdot t = b.$$

Left cancellation and conicality of T make this a partial order. Its graph is an internal hereditarily symmetric object; its definition does not require treating the generic filter as an internal parameter.

4.1 A partial order with no linear extension

Audited theorem. The native partial order on A has no linear extension in W . Consequently

$$W \models \neg\text{OEP} + \neg\text{BPI},$$

where OEP says that every partial order has a linear extension and BPI says that every nontrivial Boolean algebra has a prime ideal. The same argument applies to the parameter variants described below. The Ordering Principle still holds: every set can be linearly ordered, but an arbitrary prescribed partial order need not be extended.

Proof sketch with the compatibility step. The source monoid has a distinguished $e \neq 1$ and μ many probes m_ξ satisfying

$$m_\xi e = e, \quad m_\xi \neq 1, \quad m_\xi T \cap m_\zeta T = eT \quad (\xi \neq \zeta).$$

It also realizes every valid finite prescription of principal-cone overlaps, with exactly the prescribed identifications. A diagram is a ground injection $q : T \rightarrow I$; write $q_t(x) = q(tx)$ and $D_q = q[T]$. Cohen conditions mention fewer than μ coordinate labels.

Suppose a name for a linear extension L is supported by D_p and (c_0, p) forces its asserted properties. Restrict c_0 to D_p . Choose q with $q_e = p$ and two probes a, b . The source local-decision lemma lets q decide their comparison after the Cohen generic is fixed. A Cohen strengthening d , restricted to D_q , therefore forces, after possibly exchanging a, b ,

$$q \Vdash A_{q(a)} <_L A_{q(b)}.$$

This fixes the diagram before choosing the following native step.

The cones $bm_\xi T \setminus eT$ are pairwise disjoint. Hence some probe $t \neq 1$ has

$$\text{supp}_I(d) \cap q[btT \setminus eT] = \emptyset.$$

Set $h = bt$. Then $he = e$, $aT \cap hT = eT$, and $b <_T h$. On three copies, indexed modulo 3, prescribe

$$(i, hx) \sim (i+1, ax) \quad (x \in T).$$

This prescription is valid. Indeed, a two-step passage through the middle copy must lie in $aT \cap hT = eT$; if $ax = hy = ez$, cancellation and $ae = he = e$ give $x = y = ez$. Thus the indicated relation is already transitive, its common root has identical parameters, and no copy collapses onto itself.

Place these copies as diagrams q_i , all with $(q_i)_e = p$, and transport d to conditions d_i by coordinate permutations fixing D_p . The d_i agree on the root. Off the root, each outgoing overlap is silent in the corresponding d_i by the choice of t . Their union is therefore a valid Cohen condition. This is the essential step: a bare swap of two diagrams would not guarantee compatible Cohen bits. The diagram merger gives a common extension of the q_i . With $x_i = A_{q_i(a)}$ and $y_i = A_{q_i(b)}$, it forces

$$x_i <_L y_i <_T x_{i+1} \quad (i \bmod 3),$$

and hence a strict six-cycle in L , a contradiction.

Forcing automorphisms here transport names and conditions. They are not asserted to be automorphisms of the actual pure ZF universe. The implication $\text{BPI} \Rightarrow \text{OEP}$ is classical propositional compactness; see Jech, *The Axiom of Choice* (1973), Example 2.3.2, p. 19. Jech's §7.2 already uses finite symmetry cycles in models with OP and without OEP. The model-specific contribution here is the silent probe cone and its exact Cohen-compatible merger.

Verification scope. The auxiliary `SixCycleAudit.lean` proves validity of this cyclic overlap using the pinned source's actual `OverlapPrescription.Valid`, exact root intersections, absence of within-copy collapse, and compatibility of copied partial assignments. Its compilation reported no `sorryAx`. It does *not* formalize the full symmetric-forcing argument or the conclusion $W \models \neg \text{BPI}$. The principal records are `NATIVE_ORDER_EXTENSION_SIX_CYCLE_PROOF.txt`, `INDEPENDENT_AUDIT_SIX_CYCLE_OEP_FAILURE.txt`, and `SIX_CYCLE_LEAN_SCOPE.txt`.

4.2 Explicit Boolean-algebra and graph witnesses

Boolean algebra. There is a nontrivial Boolean algebra B_{ord} of cardinality exactly $|A|$ with no prime ideal. Use variables p_{ab} , $a, b \in A$, and impose

$$\neg p_{aa}, \quad p_{ab} \vee p_{ba} \ (a \neq b), \quad \neg(p_{ab} \wedge p_{ba}) \ (a \neq b), \quad (p_{ab} \wedge p_{bc}) \Rightarrow p_{ac}, \quad p_{ab} \ (a <_T b).$$

Take the free Boolean algebra modulo the ideal generated by the negations of these constraints. Every finite subtheory is satisfiable: linearly extend the native partial order on its finitely many mentioned points. Thus the quotient is nontrivial. A prime ideal would give a homomorphism to 2, hence a forbidden native linear extension.

Finite terms have cardinality A , so A surjects onto B_{ord} . Every well-orderable nontrivial Boolean algebra has a prime ideal in ZF: recursively include the next enumerated element in a growing proper ideal whenever this preserves properness. If it cannot be included, its complement already belongs to the ideal. Limit unions stay proper. Consequently B_{ord} is not well-orderable, and $Q(A)$ gives $B_{\text{ord}} \simeq A$. It has the least possible injection-cardinality of any prime-ideal-free nontrivial Boolean algebra. In particular, for every set X in this model,

$$X \text{ is well-orderable} \iff \text{every nontrivial Boolean algebra } B \preceq X \text{ has a prime ideal.}$$

Finite-cone realization and an unavoidable new subset. Put $S = [A]^{<\omega}$, $C_F = \{E \in S : F \subseteq E\}$, and

$$\mathcal{I} = \{D \subseteq S : (\exists F \in S) D \cap C_F = \emptyset\}.$$

Canonically linearly extend each finite native order on E by repeatedly taking its lexicographically least remaining minimal point; call the result L_E . The old sets

$$D_{ab} = \{E \in S : a, b \in E \text{ and } a <_{L_E} b\}$$

satisfy the order constraints modulo $\mathcal{I}$. The subalgebra generated by their classes inside $\mathcal{P}(S)/\mathcal{I}$ is another size- A algebra with no prime ideal. The entire quotient need not have size A .

Even an ultrafilter on the *old* algebra $\mathcal{P}(S)^W$ containing all C_F defines a native linear extension by $a <_U b \iff D_{ab} \in U$. Thus forcing with the positive part of $\mathcal{P}(S)^W/\mathcal{I}$ necessarily adds a new subset of A^2 , and therefore of A . Moreover, the A^2 -indexed dense open sets deciding the Boolean values $[D_{ab}]$ have empty intersection. In the Boolean completion they witness failure of binary distributivity indexed by A . No claim about countable distributivity follows. Genericity decides old subsets; it does not automatically give an ultrafilter on the enlarged final powerset.

Graph compactness. There is a graph G with $|G| = |A|$ such that every finite subgraph is 3-colorable but $\chi(G) = 4$. The order theory above is a finitely satisfiable 3-CNF theory after repeating literals. Encode Boolean variables by opposite vertices adjacent to a fixed base vertex in a triangle of colors T, F, B . A clause $x \vee y \vee z$ is equivalent to

$$\exists w (\text{NAE}(x, y, w) \wedge \text{NAE}(\neg w, z, F)),$$

where NAE means “not all equal.” Each NAE constraint is represented by a fresh triangle whose three vertices are respectively adjacent to the three Boolean terminals. It has an extending 3-coloring exactly when NAE holds. A global 3-coloring would solve the impossible order theory. Every finite subgraph is colorable by finite satisfiability, while a canonical 4-coloring assigns Boolean values uniformly and colors each gadget greedily. Tagged finite powers of A give the exact vertex cardinality.

Well-orderable graphs satisfy finite-color compactness in ZF by transfinite recursion, choosing at each vertex the first color preserving finite satisfiability. Thus A is again a least possible counterexample cardinal. In contrast, 2-color compactness holds in W : finite bipartiteness gives a canonical pair of parity classes in each nontrivial connected component, and AC_2 selects one class per component. Here AC_2 follows from OP. For each fixed finite $k \geq 3$, joining this graph with a complete graph on $k - 3$ fresh vertices gives a graph whose finite subgraphs are k -colorable while its chromatic number is $k + 1$. Thus finite- k compactness fails for every such k . The general compactness connection is classical; see de Bruijn–Erdős (1951) and Cowen–Hechler–Mihok (2002). The explicit gadgets and infinite argument are checked in `EXPLICIT_GRAPH_COMPACTNESS_INDEPENDENT_AUDIT.txt`; the Boolean proof is in `ORDER_OBSTRUCTION_BOOLEAN_ALGEBRA_AND_FINE_FORCING.txt`.

4.3 BPI, small violations of choice, and limits of compactness

Under $\text{SVC}(A)$, BPI is equivalent to the existence of a fine ultrafilter on $[A]^{<\omega}$, equivalently a prime ideal in $\mathcal{P}([A]^{<\omega})/\mathcal{I}$. This is the Pincus–Blass local reflection recorded in Ryan-Smith, *Local reflections of choice*, Proposition 3.13. The preceding theorem settles this particular local test negatively in W ; it does not prove the abstract implication $\text{SVC}(A) + \text{AC}_{\text{WO}} + Q(A) + \text{BPI} \Rightarrow \text{AC}$.

The model gives relative nonimplications

$$\text{PP} \not\Rightarrow \text{BPI}, \quad \text{NDS} \not\Rightarrow \text{BPI}, \quad \text{FB} \not\Rightarrow \text{BPI},$$

and likewise with OEP as conclusion. In the reverse direction, the classical basic Cohen model satisfies BPI and SVC and has an infinite Dedekind-finite set. It fails PP, NDS, FB, and Ramsey’s principle on arbitrary infinite sets. This is an old model, not a new construction; the SVC generator need not be its Cohen set. Tachtsis’s 2024 Theorem 6.4 supplies the stronger

opposite-direction example $\text{BPI} + \text{AC}_\omega + \neg\text{NDS}$, hence also $\neg\text{FB}$ and ordinary infinite homogeneity. Its exact attribution was verified through Blass–Kulshreshtha’s primary manuscript, Theorem 10.5; the full Tachtsis article was not obtained. These statements do not decide whether PP implies NDS in arbitrary ZF models, or whether PP and BPI can coexist with $\neg\text{AC}$.

Several useful subsidiary facts survived the unsuccessful positive-BPI routes. FB gives every proper filter a member of minimum injection cardinality; restricting an ultrafilter to that member makes it cardinal-uniform. This is not a cofinal filter base or an ultrafilter existence theorem. The native lower cones generate a proper filter closed under all internal well-ordered intersections and avoiding every well-orderable subset of A . An ultrafilter extending it need not retain that completeness. In the original model every nonprincipal ultrafilter on $A \subseteq \mathcal{P}(\omega_1)$ is countably incomplete: first push forward by the first disagreement with its coordinatewise ultrafilter decisions, obtaining a nonprincipal image on ω_1 ; then use the ground injection $\omega_1 \rightarrow \mathcal{P}(\omega)$ and repeat, obtaining a nonprincipal image on ω .

Likewise, BPI and PP give a valid small-carrier Henkin lemma: a finitely satisfiable finitary theory whose symbols inject into A has a model of cardinality at most A , provided $A^{<\omega} \simeq A$. This does not transport the original monoid action through an arbitrary bijection of carriers. Finite-factor ultraproduct arguments also do not force strict external cardinal descent: in ZFC every unbounded finite product $\prod_n k(n)/U$, for nonprincipal U on ω , has size $2^{\aleph_0}$, even after finitely many iterated logarithms of the factors. These boundaries are recorded in `STRUCTURAL_BPI_PHASE1.txt`, `ANCESTOR_FILTER_COMPACTNESS_AND_ULTRAFILTER_INCOMPLETENESS.txt`, `HENKIN_SMALL_CARRIER_AND_STRUCTURE_OBSTRUCTION.txt`, and `ULTRAPRODUCT_AND_QUOTIENT_OBSTRUCTIONS.txt`.

4.4 Parameter changes and preservation

Audited parameter theorem. Let the ground satisfy ZFC. Suppose κ is regular uncountable, μ is regular, and

$$2^{<\kappa} < \mu, \quad \lambda^{<\kappa} < \mu \quad \text{for every infinite ground cardinal } \lambda < \mu.$$

Use the source monoid of size μ , $|I| = \mu^+$, Cohen conditions of size $< \kappa$, and diagram supports of size at most μ . The audited adaptation yields the same SVC, ordinal-choice and quotient package, hence PP, FB, NDS and $\neg\text{AC}$. The quotient proof requires an explicit repair: bounded arrays indexed below ω_1 become arrays of size $< \kappa$, controlled by the strengthened no-new-ground-sequences lemma. The countable germ construction remains countable. The native six-cycle and real-height obstruction remain valid, giving $\neg\text{BPI}$ and $\neg\text{DC}_{\omega_1}$ in every variant, although ordinary DC holds.

Two internal thresholds are distinct. If W_λ says that every set is injection-comparable with the well-ordered cardinal λ , then

$$w(A) = \mu, \quad h(A) = \mu^+, \quad W_\lambda \iff \lambda \leq \mu, \quad \mathcal{P}(\alpha) \text{ is well-orderable} \iff \alpha < \kappa.$$

Here $w(A)$ denotes the largest well-ordered cardinal injecting into A . For $\alpha < \kappa$, the entire powerset equals its ground counterpart. Under $V = L$, write M_{mn} for $\kappa = \aleph_m$, $\mu = \aleph_n$, $1 \leq m < n < \omega$. All ground alephs are preserved:

Model	κ	μ	$W_{\aleph_3}$	$\mathcal{P}(\aleph_1)$ well-orderable
M_{12}	$\aleph_1$	$\aleph_2$	no	no
M_{13}	$\aleph_1$	$\aleph_3$	yes	no
M_{23}	$\aleph_2$	$\aleph_3$	yes	yes

These are specific separations over a common strong package, not unrestricted independent assignments of two sentences. The consistency interpretation uses the source’s formula-by-formula

forcing method over L ; ordinary $\text{Con}(\text{ZF})$ suffices, with no inaccessible cardinal. The complete adaptation is not Lean-formalized. See `LARGER_MU_SECOND_INDEPENDENT_AUDIT.txt`.

Without GCH, put $\theta = (2^{<\kappa})^V$. The ground Cohen forcing is θ^+ -cc and preserves κ and its regularity. Every old binary string of length $< \kappa$ occurs as a block of any one generic row. The row's block map therefore collapses the old set of such strings, of size θ , to cardinality κ internally in W . Thus

$$(\kappa^+)^W = (\theta^+)^V, \quad (\kappa^{<\kappa})^W = \kappa.$$

Ordinal-function locality transfers the relevant preservation from the Cohen extension, not from the full outer product extension. The exact no-GCH pair-coloring argument respects this identification; see `NO_GCH_HOMOGENEOUS_BOUND_RECHECK.txt`.

Preservation theorem. If $W \subseteq N$ have the same ordinals, N adds no subsets of A , and N satisfies $\text{SVC}(A)$, then N retains $\text{AC}_{\text{WO}} + Q(A) + \neg \text{AC}$ and its PP/FB consequences. Indeed, $A^2 \simeq A$ makes every new map on A have an old kernel. An old non-well-orderable quotient cannot become well-orderable without well-ordering A . Old ordinal-choice then supplies representatives for kernels of new ordinal-valued maps. SVC reduces general ordinal-choice to these maps. Ordinary set forcing always preserves SVC by evaluating a ground presentation of the set of subnames.

This criterion has concrete instances. If the outer ZFC forcing has size χ , take regular $\lambda > \chi$ and the ground forcing $\text{Add}(\lambda, \delta)^V$. Easton's small-forcing argument makes it $< \lambda$ -distributive in the outer extension. The audited intersection lemma $W[K] \cap V[F][H] = W$ then gives no new subsets of A^2 when λ exceeds its outer size. Taking λ also above the outer size of $\mathcal{P}([A]^{<\omega})^W$ preserves the BPI truth value; here that means preserving its failure. For the original L -ground model one may use $\lambda = \aleph_5^V$ and $\delta \geq \lambda^+$ with $\delta^\lambda = \delta$. The largest well-ordered cardinal injecting into $\mathcal{P}(\lambda)$ becomes δ , while that powerset stays non-well-orderable. These are arithmetic variations, not further weak-choice separations by themselves. See `HIGH_FORCING_PRESERVATION_INDEPENDENT_AUDIT.txt`.

4.5 Exact pair and finite-arity Ramsey spectra

For an infinite well-ordered cardinal λ , use the temporary notation $\text{NR}_\lambda^{r, \text{fin}}$ for: every finite coloring of $[X]^r$, for every non-well-orderable X , has a homogeneous subset of cardinality λ . The requirement concerns all sets and colorings, not only a distinguished witness. Under AC it is vacuous. Ordinary Ramsey homogeneity with an unspecified infinite size already follows from DC here.

Pairs, without GCH. Throughout the audited family, the largest λ for binary pair colorings is exactly κ . More strongly, every pair coloring with an ordinal color cardinal $1 \leq \eta < \kappa$ on a non-well-orderable set has a homogeneous subset of size κ . At $\eta = \kappa$, the coloring

$$c(\{a, b\}) = \min(a \triangle b), \quad a, b \in A,$$

has no homogeneous triple: three binary bits cannot be pairwise different at the same first-difference coordinate. Thus κ is the exact least well-ordered color cardinal permitting a failure of infinite pair homogeneity on a non-well-orderable set.

For the positive direction, encode a pulled-back coloring on κ^+ , the enumeration of $\kappa^{<\kappa}$, and ordinal-sized injection witnesses into a set C of ordinals. The inner model $L[C]$ computes the required cardinals and $2^{<\kappa} = \kappa$ correctly. Apply the Erdős–Rado theorem there, in the form $(2^{<\kappa})^+ \rightarrow (\kappa + 1)_\eta^2$ for $\eta < \kappa$. The resulting homogeneous set belongs to W . Use $\kappa^+ \preceq A$ and then the copy of A in every non-well-orderable set. The precise primary source is Baumgartner–Hajnal–Todorćević, *Extensions of the Erdős–Rado Theorem*, Theorem 2.1, p. 4.

For the negative direction, color $\{a, b\}$ by the parity of their first difference. Fresh-copy locality first makes every homogeneous subset well-orderable. A Cohen Δ -system argument

excludes size κ^+ : thin conditions to equal row patterns and then prescribe a fresh first difference of the opposite parity. Both colors have homogeneous subsets of size κ , supplied by rows whose first 1 occurs at increasing coordinates of a fixed parity. In the no-GCH case the argument uses the preserved ground θ^+ , which is the internal κ^+ . Consequently

$$\text{NR}_\lambda^{2,\text{fin}} \iff \lambda \leq \kappa.$$

Sources and complete proofs are `EXACT_HOMOGENEOUS_CARDINAL_BOUND.txt` and `GLOBAL_NON_WELL_ORDERABLE_RAMSEY_THRESHOLD.txt`.

Higher arities over L . In M_{mn} the exact threshold for all finite color counts is

$$\max\{\lambda : \text{NR}_\lambda^{r,\text{fin}}\} = \aleph_{\max(m-r+2,0)}, \quad 2 \leq r < \omega.$$

Model	pairs	triples	4-tuples	5-tuples
M_{12}, M_{13}	$\aleph_1$	$\aleph_0$	$\aleph_0$	$\aleph_0$
M_{23}	$\aleph_2$	$\aleph_1$	$\aleph_0$	$\aleph_0$
M_{34}	$\aleph_3$	$\aleph_2$	$\aleph_1$	$\aleph_0$

The positive proof applies finite-arity Erdős–Rado inside $L[C]$; all powersets needed for its finite GCH calculation lie strictly below κ and are unchanged. The negative proof begins with Sierpiński’s comparison of a ground ordinal enumeration and binary lexicographic order, then steps up using consecutive first differences. A tuple is colored by the preceding color together with an increasing/decreasing tag, or by an additional nonmonotone color. Finite Ramsey excludes an infinite homogeneous set of the latter color. DC makes a homogeneous set in either remaining orientation well-ordered by lex or reverse lex; its first differences yield a forbidden homogeneous set for the prior coloring. This avoids an auxiliary well-order of $\mathcal{P}(\kappa)$. The negative triple witness uses four colors; successive constructions use at most $5 \cdot 2^{r-3} - 1$ colors, not claimed optimal. See `HIGHER_ARITY_ROOT_INDEPENDENT_AUDIT.txt` and Erdős–Rado (1956), Theorem 39, pp. 467–469.

Binary triples remain a separate question. The saved bounds are

	proved positive	unresolved here	proved negative
M_{12}, M_{13}	ω	ω_1	ω_2
M_{23}	ω_1	ω_2	ω_3

for homogeneous cardinalities of arbitrary binary triple colorings on non-well-orderable domains. The binary orientation coloring of adjacent first differences has exact largest homogeneous cardinality κ , in both colors; this gives the stated negative upper bounds, not a universal positive theorem at κ . Every one of the sixteen maps merging the particular four-color triple witness into two colors has an ω_1 -sized homogeneous set in M_{12} and M_{13} . Projections use row patterns; the XOR case uses a countable binary tree of coordinates labeled by nested real intervals. Thus simple color merging cannot settle the missing negative result. The classical binary stepping-up proof uses an auxiliary well-order unavailable internally. See `BINARY_TRIPLE_ROOT_INDEPENDENT_AUDIT.txt`. No claim is made that the unresolved sentences are open in the entire literature.

4.6 No-go results for repairing the model

Increasing $|I|$ alone cannot produce BPI while retaining the absorption probes, all valid three-copy mergers, root-fixing transport, local decision, and Cohen conditions mentioning fewer than μ labels. Those ingredients already prove the six-cycle obstruction. More abstractly, the literal infinite-product version of finite Ramsey homogeneity is false: parity relative to representatives modulo finite coordinate disagreement gives a binary coloring with no monochromatic rectangle having infinitely many nonsingleton factors.

Two concrete order-based changes were audited. Replacing the full coordinate group by automorphisms of a saturated dense linear order adds a supported cut map $A \rightarrow \mu^+$, while support locality still forbids $\mu^+ \hookrightarrow A$. Thus PP and AC_{WO} fail. Alternatively, decorate every entire diagram with a ground linear extension of T , preserving that order under restriction. The root-pair amalgamation survives, but the internal profile map surjects onto all ground linear extensions of T , a set of ground size 2^μ . Again PP and ordinal-choice fail. This differs from finite order forcing over W , discussed elsewhere in this report, whose orbit profiles are new and need not be ground. See `ORDERED_SUBGROUP_CUT_QUOTIENT_OBSTRUCTION.txt` and `ORDER_REPAIR_ROOT_AUDIT.txt`.

For restricted palettes $\mathcal{C}$ of orders on T , closed under $\vartheta \mapsto \vartheta_t$ with $x \vartheta_t y \iff tx \vartheta ty$, there is no nonempty finite palette, even without requiring extension of native divisibility. Finite Ramsey produces four probes monotone up to reversal in every palette order; a five-chart tournament merger would give a finite set with no least element. For infinite $|\mathcal{C}| = \lambda$, the same argument and Erdős–Rado give the necessary bound $\mu \leq 2^{2^\lambda}$. It is not an optimal bound. Palettes of size at most μ can satisfy pullback closure and every acyclic finite request. A universal lift can also be constructed for one prescribed palette, using cross-denominator probe cones sharing exactly the root. The unresolved issue is closure of one final palette under universal lifts for all its own witness types, followed by support descent and quotient classification. No positive-BPI model or universal impossibility theorem results. See `ORDER_PALETTE_ROOT_AUDIT.txt`.

Ransom’s 2026 dissertation gives an exact forcing-aware criterion for BPI (Definition 4.3.1 and Theorem 4.3.3, pp. 93–99), involving dense sets, bases for the normal filter, orbit colorings, and compatibility below all common extensions of finitely many translated conditions. The criterion concerns the full Cohen–diagram product. A Ramsey property of the abstract monoid alone does not verify it. Its finite-support examples do not automatically cover these μ -sized supports. The six-cycle theorem now rules out a successful verification for the unchanged model. The earlier precise compatibility obligations remain documented in `SYMMETRIC_FILTER_PHASE3.txt` and `SYMMETRIC_FILTER_PHASE4_LARGE_I.txt`.

4.7 Source corrections, attribution, and remaining boundaries

The implication $\text{AC}_{\text{WO}} \Rightarrow \text{DC}$ is valid in pure ZF: Jech, Theorem 8.2, pp. 121–122, attributes it to Jensen. An early apparent contrary table entry was an erroneous HTML reconstruction, corrected by inspecting the actual primary PDF and its prose. Here ordinal-choice means choice for well-orderable *index families*, not arbitrary families of well-orderable members. The verified chain is $\text{PP} \Rightarrow \text{AC}_{\text{WO}} \Rightarrow \text{DC}$; it does not yield DC_{ω_1} .

A different source conflict remains recorded rather than resolved by recency. Pincus’s original announcement and publisher summary, Ransom’s 2025 paper, and Asperó–Karagila’s 2020 manuscript explicitly attribute models of $\text{ZF} + \text{DC} + \text{BPI} + \neg \text{AC}$ to Pincus. Kowalczyk–Zapletal, arXiv:2607.16747v1, instead calls the corresponding implication to AC open; its actual theorem establishes a restricted compactness fragment. The complete relevant Pincus proof was not retrieved in this investigation. This discrepancy is not used as a premise, and no claim that a positive BPI variant would solve that historical problem is made. See `DC_BPI_SOURCE_CONFLICT.txt`.

The bounded novelty check found no BPI or order-extension statement in the twelve text files of the pinned PP manuscript and no prior assertion of this particular PP/FB/NDS countermodel to BPI in the sources checked. That is limited search evidence, not a priority certificate. The classical compactness implications, symmetry-cycle method, Ramsey theorems and coloring constructions retain their established attributions. The source construction, not this report, proves the foundational PP consistency theorem. The additional results here are model consequences and explicitly audited adaptations.

Primary references used in this section. Stable source links are: Jech (1973); Felgner–Truss (1999), OEP and BPI; Ryan-Smith, Proposition 3.13; Ransom (2025) and Ransom’s 2026 dissertation; Tachtsis (2024) and Blass–Kulshreshtha, version 3; Baumgartner–Hajnal–Todorćević, Theorem 2.1; Erdős–Rado (1956); Erdős–Hajnal–Rado (1965), Lemma 5A; and Cowen–Hechler–Mihok (2002). All named local audit files belong to the accompanying `evidence/original_mode 1/` archive directory (mapped from the original `BPI_RPP_five_hour_research/` record). Their original intermediate statuses are preserved; the later independent audits and this consolidation specify which claims supersede them.

5 The ordinary extension by finite linear orders

This section reports the second model studied in the workspace. Starting with the OpenAI symmetric model W , we force a linear extension of its native partial order and obtain $N = W[G]$. The established conclusions include

$$N \models \text{ZF} + \text{AC}_{\text{WO}} + \text{DC} + \neg\text{AC} + \neg\text{DC}_{\omega_1}, \quad \text{SVC}(A) + \text{SVC}(\mathbb{R}_N),$$

where $\mathbb{R}_N = \mathcal{P}(\omega)^N$. All old cardinal comparisons survive, but the new real set is not well-orderable. Whether N satisfies *global* PP, NDS, FB, or BPI remains unresolved. In particular, this section does not present a model of BPI together with failure of Choice.

Here $X \approx Y$ means equipotence, $X \preceq Y$ means the existence of an injection, and “old” means belonging to W , rather than merely to the earlier ZFC construction ground V . A result described as audited has a written mathematical proof and a separate workspace check; this is neither external peer review nor a formal verification of the whole forcing construction. The proofs below are summaries of the saved arguments. The audit register at the end supplies their exact filenames.

5.1 Construction and the chain conditions

The starting model has a non-well-orderable set A and a ground monoid T of well-orderable cardinality μ . There is an internal free right action $(a, t) \mapsto a \cdot t$ of T on A . “Free” here means that $t \mapsto a \cdot t$ is injective for each fixed a ; it does not mean that right multiplication $a \mapsto a \cdot t$ is injective. Define

$$a \leq_T b \iff (\exists t \in T) b = a \cdot t.$$

This is a partial order. Its principal upset aT is called a *chart*; it is an old well-orderable set of cardinality μ . Every old well-orderable subset of A lies in one chart. The results used from W include

$$A^2 \approx A, \quad A \times \mu \approx A, \quad (A^\omega)^W \approx A, \quad h^W(A) = \mu^+.$$

Every old set has a normal form $(A \times \alpha) \sqcup \beta$, with ordinal coefficients. Here $h(X)$ is the least ordinal not injecting into X ; $w(A) = \mu$ means that μ is the largest well-orderable cardinal represented by a subset of A .

Let $\mathbb{P}_{\text{ext}}$ consist of finite strict linear orders on finite subsets of A which extend the strict part $<_T$ of the native order on their domains. A stronger condition extends the domain and the finite order. The generic union $<_L$ is a strict linear order on all of A extending $<_T$; we also write L for this order. Set

$$N = W[G], \quad G \subseteq \mathbb{P}_{\text{ext}}.$$

This is ordinary set forcing over the choiceless model W . Its construction and its preservation properties require separate proofs.

Audited chain-condition theorem. Every antichain in $\mathbb{P}_{\text{ext}}$ is countable, and every predense subset has a countable predense subset. We call these properties CCC_2 and CCC_3 , respectively. The second is not inferred from the first in arbitrary ZF models.

The local proof uses the source monoid’s height $\rho : T \rightarrow (0, 1)$, positive costs ϵ_t , and conditional joins. Two conditions $(x_0 < \dots < x_{n-1})$ and $(y_0 < \dots < y_{m-1})$ are incompatible exactly when a reversed rectangle occurs:

$$i < j, \quad l < k, \quad x_j \leq_T y_l, \quad y_k \leq_T x_i.$$

After bounding the finite lengths and bounding the costs away from zero, finite Ramsey theory turns a sufficiently large antichain into a triangular pattern. Successive finite joins would then increase ρ by more than a fixed positive amount at every step, contradicting $0 < \rho < 1$. This proves a local σ -bounded chain condition on a chart. An alleged ω_1 -indexed global antichain localizes to one chart. In W every uncountable set contains ω_1 , so this excludes all uncountable antichains.

Each chart subforcing is regular in $\mathbb{P}_{\text{ext}}$. A finite condition has a finite reduction to the chart: its constraints are projected to the least native successors in that chart, using conditional joins. The stronger CCC_3 proof then uses the actual source construction’s petal-separation lemma. If a predense set has supporting diagram p , choose a larger diagram q with $q_e = p$. Its part inside the chart of q is globally predense. A greedy well-ordered construction in that chart produces a countable predense subfamily. The source-specific petal and gate calculations are material hypotheses of this argument [N1].

5.2 Preservation and the remaining failures of Choice

Audited preservation theorem. The extension preserves all ordinal cardinals and cofinalities, the Hartogs number of every old set, and the non-well-orderability of every old non-well-orderable set. More precisely, every new well-orderable subset of an old set X is contained in an *old* well-orderable subset of X [N2].

For the proof let $\mathbb{B} = \text{RO}^W(\mathbb{P}_{\text{ext}})$. Its antichains are countable: an uncountable Boolean antichain would contain an ω_1 -indexed subfamily, and old AC_{WO} would select incompatible forcing conditions below its members. If a name is forced below b_0 to be a function $f : X \rightarrow Y$ between old sets, put

$$g(x) = \{y \in Y : b_0 \wedge \|\dot{f}(\check{x}) = \check{y}\| > 0\}.$$

For fixed x the nonzero Boolean values are disjoint. Thus $g(x)$ is countable in W , and g is one canonically defined old function with $f(x) \in g(x)$. For an ordinal domain, old AC_{WO} well-orders the union of these possible-value sets. This gives the asserted covering property, from which the cardinal, cofinality and Hartogs conclusions follow. In particular

$$w^N(A) = \mu, \quad h^N(A) = \mu^+, \quad N \models \neg \text{AC}.$$

The complete-Boolean-algebra covering argument is related to the choiceless chain-condition analysis of Karagila–Schweber; the saved proof checks the extra hypotheses needed here, instead of importing ZFC preservation wholesale.¹

The stronger chain condition separately preserves AC_{WO} . For each member of a new ordinal-indexed family, use a countable predense set of conditions carrying witness names. Collection first bounds those names in one old set. Old ordinal-indexed choice selects the countable lists and witnesses; in the generic extension, the first successful condition supplies the chosen element.

¹A. Karagila and N. Schweber, “Choiceless chain conditions,” *European Journal of Mathematics* 8, Supplement 2 (2022), S393–S410, <https://doi.org/10.1007/s40879-022-00564-2>.

This uses local witness density, not the choiceless maximum principle. Consequently N also satisfies ordinary DC, by the established implication $\text{AC}_{\text{WO}} \Rightarrow \text{DC}$ [N2].

Every new ordinal sequence of A -points is contained in one old chart. It therefore has a strict common native predecessor. A descending ω_1 -sequence would produce, in that chart, a strictly decreasing ω_1 -sequence of real heights, which is impossible. The resulting countably closed tree of descending native sequences witnesses $\neg\text{DC}_{\omega_1}$. Also L has no well-orderable coinital subset. Its reverse witnesses failure of the Cofinality Principle, the assertion that every linear order has a well-orderable cofinal subset. These statements concern new subsets as well as old ones.

The old Boolean algebra $\mathbb{B}$ has cardinal A , is complete, atomless and ccc in W , and has no ultrafilter there: an ultrafilter would evaluate its comparison values to a forbidden native linear extension. The generic extension supplies an ultrafilter on this *old* algebra. Its old completeness need not extend to all new families of Boolean elements. Repairing this particular obstruction does not establish BPI for all Boolean algebras of N [N3].

5.3 The new real set and the unresolved evaluation quotient

Audited real-cardinal theorem. In N there are maps

$$A \hookrightarrow \mathbb{R}_N, \quad A \twoheadrightarrow \mathbb{R}_N.$$

The real set is not well-orderable and satisfies

$$w(\mathbb{R}_N) = \mu, \quad h(\mathbb{R}_N) = \mu^+, \quad \ell(\mathbb{R}_N) = \mu^+, \quad (A^\omega)^N \approx \mathbb{R}_N.$$

For the original parameters $\mu = \aleph_2$, this gives the non-well-orderable real set Hartogs number $\aleph_3$; parameter variants use their preserved μ . Here ℓ is the least positive ordinal not obtainable as a surjective image; the harmless empty-set convention does not affect the displayed infinite value [N4].

For the injection, fix selected probes m_n and encode the order pattern of $\{a \cdot m_n : n < \omega\}$ as a real r_a . The probe geometry implies that the sets at two distinct centers intersect in at most one point. A finite-order swap can change a comparison at one center while keeping the corresponding comparison at the other unchanged. These separating extensions are dense, so $a \mapsto r_a$ is injective.

For the surjection, CCC_3 expresses every old Boolean value as a countable join of forcing conditions. Since $|\mathbb{P}_{\text{ext}}| = |A|$ and $(A^\omega)^W \approx A$, one old A -sized set codes all old sequences of Boolean values. Evaluation under the generic ultrafilter covers every new subset of ω . This produces a surjection, not an injection in the opposite direction.

If μ^+ injected into $\mathbb{R}_N$, new AC_{WO} would choose preimages under $A \twoheadrightarrow \mathbb{R}_N$, contradicting the preserved Hartogs number of A . This proves the upper threshold; the profile injection gives the lower bound. Finally, $\mathbb{R}_N \preceq A^\omega \preceq \mathbb{R}_N^\omega \approx \mathbb{R}_N$ proves the countable-power identity. The left-hand power here is computed in N , not in W .

The central unresolved comparison is

$$\mathbb{R}_N \preceq A ?$$

An injection and a surjection from A to the same target do not settle this in ZF. Least choices of ground code *types* do not choose centers in the remaining non-well-orderable fibers. Even $\mathbb{R}_N \approx A$, if eventually proved, would settle one quotient instance and would not alone prove global PP.

5.4 Local real fields and exact localization

Put $Q = \mathbb{P}_{\text{ext}}(T)^V$. For each $a \in A$, the chart map $t \mapsto a \cdot t$ pulls the generic order back to a Q -generic filter K_a . Chart regularity gives genericity over W , hence over V ; it does not justify genericity over the external Cohen universe $V[F]$. Fix in V a μ -enumerated set Γ of all countable Q -names coding reals, and define internally

$$R_a = \{\tau^{K_a} : \tau \in \Gamma\}.$$

Externally this is $\mathbb{R}_N \cap V[K_a]$. The internal definition uses a set of names, not a predicate for the ground universe.

Audited local-field theorem. For every a , $|R_a| = \mu$. Every new well-orderable subset of $\mathbb{R}_N$ is contained in some R_a . Every new subset of an ordinal belongs to $W[K_a]$ for some a . Moreover

$$R_a \subseteq R_b \iff a \in bT, \quad r_a \in R_b \iff a \in bT.$$

Indeed the stronger exclusion $r_a \notin W[K_b]$ holds whenever $a \notin bT$ [N5].

The size bound comes from the ground enumeration of Γ ; the μ profiles $r_{a \cdot t}$ give equality. For localization, take the Boolean bits of an ordinal-indexed real family and choose their countable forcing codes using old ordinal-indexed choice. All finite point domains form one old well-orderable subset of A , hence lie in one chart. Each individual real then has a countable sequence of ground Q -conditions. The original model adds no countable sequences of ground elements, so that code belongs to V and evaluates in $V[K_a]$. The full uncountable array need only belong to W ; this explains the weaker $W[K_a]$ conclusion for arbitrary ordinal subsets.

The inclusion-reflection proof uses a quotient-forcing argument. If $a \notin bT$, infinitely many of the probes at a have identical native comparison profiles to the entire chart bT . Below any finite quotient condition, two remain interchangeable while all chart-order information is fixed. Both orientations can still be forced, so the corresponding bit of r_a is new over $W[K_b]$.

Two distinctions are essential. A well-orderable real family is *contained in* R_a ; the family itself need not belong to $V[K_a]$. Also proper field inclusions do not give strict cardinal inequalities: all the R_a have the same cardinal μ . Every well-orderable family of these fields is contained in one strictly larger field, by taking a strict native predecessor of its centers. Thus there is no well-orderable cofinal subfamily under inclusion.

5.5 All comparisons between old sets are preserved

For any old X, Y , the existence in N of an injection, bijection, or surjection between them is equivalent to its existence in W . Every new surjection between old sets has the PP conclusion witnessed by an *old* injection in the reverse direction. Moreover every new set-family consisting of old sets has a finite coinital cardinal basis [N6].

Here is the comparison invariant. An old well-orderable set of cardinality ν has record $(0, \nu)$. Normalize a non-well-orderable old set as

$$X \approx (A \times \alpha) \sqcup \beta, \quad \alpha \geq \mu, \quad \beta = 0 \text{ or } \beta > \alpha,$$

with ordinal cardinals as coefficients, and give it record (α, ε) , where $\varepsilon = \max(\alpha, \beta)$. These records are unique and injection comparison is componentwise, both in W and in N .

For example, an injection from record (α, ε) into (γ, ζ) gives $\varepsilon \leq \zeta$ by Hartogs preservation. Each of the α source copies of A must meet the target's A -part; otherwise that copy would become well-orderable. New ordinal-indexed choice selects one point from each such disjoint

intersection, giving $\alpha \leq \gamma$. Conversely the two inequalities already construct an old injection. A similar preimage selection proves preservation of surjections.

Every set of ordinal pairs has a finite coinital subset under the componentwise order: repeatedly choose a least first coordinate and then a least second coordinate not already dominated. Continuing infinitely would strictly decrease the second coordinate infinitely. Applying this to the unique old records proves the relative FB claim. For a new family of old sets, first bound their ranks inside one old rank segment, so its old record assignment is a set function.

Thus an infinite new cardinal antichain or descending chain cannot consist solely of old cardinalities. This is not global FB or NDS for all new sets. If $\mathbb{R}_N \not\approx A$, its cardinality is new: A is the only old non-well-orderable cardinality with $w = \mu$.

5.6 Explicit nonsplitting and additional choice bounds

For every $t \neq 1$, the old surjection $f_t : A \rightarrow A$, $f_t(x) = x \cdot t$, remains without a right inverse. More strongly, there is no assignment of a nonempty countable subset of each fiber, even in N [N7]. The old proof says that any countable-valued assignment on A has some value contained in the forward chart of its argument. A nonempty value consisting of predecessors would then contradict conicality of T . For a new assignment, CCC_3 supplies a canonical old countable cover of every possible value, reducing the contradiction to that old statement.

Transport f_t along $a \mapsto r_a$ and extend it by the identity outside that image. This gives a surjection $g_t : \mathbb{R}_N \rightarrow \mathbb{R}_N$ without a right inverse, or even a countable multisection. Consequently choice fails for real-indexed families of nonempty sets of reals. This is not a failure of PP: PP does not split each specified surjection, and these two sets are already equipotent.

Ordinary forcing preserves $\text{SVC}(A)$. The profile injection gives a retraction $\mathbb{R}_N \twoheadrightarrow A$, hence $\text{SVC}(\mathbb{R}_N)$. Coding $\mathbb{R}_N \times \eta$ into a power set of an ordinal yields

$$\text{KWP}_1^*, \quad \text{KWP}_2 \quad \text{in } N.$$

Here KWP_n means that every set injects into an n -fold power set of an ordinal; the starred version uses partial surjections. Inverse-image fibers give $\text{KWP}_1^* \Rightarrow \text{KWP}_2$. These are the established bounds, not KWP_1 , the Ordering Principle, or global finite-choice preservation.²

There are also definite negative continuum-function conclusions. Write ICF for $\mathcal{P}(X) \approx \mathcal{P}(Y) \Rightarrow X \approx Y$, and HCF for reflection of injection comparison by $\mathcal{P}$. ZF proves $\text{ICF} + \text{KWP}_{n+1} \Rightarrow \text{KWP}_n$: apply the latter to $\mathcal{P}(X)$, obtaining $\mathcal{P}(X) \preccurlyeq \mathcal{P}(B)$ with $B = \mathcal{P}^n(\theta)$ and θ infinite. Since B absorbs two copies, $\mathcal{P}(X \sqcup B) \approx \mathcal{P}(B)$, so ICF gives $X \preccurlyeq B$. Finite descent reaches AC. Thus the proved $\text{KWP}_2 + \neg \text{AC}$ package implies $\neg \text{ICF} + \neg \text{HCF}$ in N [N8].

5.7 Initial segments, final segments, and open intervals

For $a <_L b$ put

$$I_a = \{x : x <_L a\}, \quad J_a = \{x : a <_L x\}, \quad (a, b)_L = \{x : a <_L x <_L b\}.$$

The final audited result is

$$I_a \approx J_a \approx (a, b)_L \approx A.$$

The earlier possibility that a final segment might be a proper square root of A was resolved by an explicit injection, not by cancelling the established identity $J_a^2 \approx A$ [N9].

²A. Karagila and J. Schilhan, “Intermediate models and Kinna–Wagner Principles,” *Proceedings of the American Mathematical Society* 154 (2026), 393–403, <https://doi.org/10.1090/proc/17425>; see Section 3.2 and Proposition 4.1 of <https://arxiv.org/html/2409.07352v2>.

An initial segment contains the old A -sized fiber $\{x : x \cdot t = a\}$ for any $t \neq 1$. For an interval, choose selected probes m_n with $m_n e = e$, pairwise cone intersection eT , and only identity common left divisors. Fix d with $d \cdot e = b$, and set $F = \{x : x \cdot m_0 = d\} \approx A$. For each $x \in F$ the ray

$$S_x = \{x \cdot m_n : n \geq 1\}$$

lies strictly below b . These rays are disjoint for different x : a shared ray point together with the shared anchor d forces the common-chart join quotients to divide both m_0 and a later m_n , so both quotients are 1.

In a finite condition, a point below b can dominate at most one point of S_x ; dominating two would place it natively above b . A remaining ray point can therefore be inserted immediately before b . Genericity makes S_x cofinal below b . Its least indexed point above a injects F into $(a, b)_L$, proving the theorem. Taking $b = a \cdot e$ also proves the final-segment case.

The whole L has countable cofinality. A fixed countable sequence of fresh merger letters with disjoint cones gives, above any center, an old countable cofinal ray. Every old non-well-orderable subset of A is nevertheless L -coinitial: a point outside a chart covering a finite condition can be placed first. Thus no final segment or bounded interval contains an old non-well-orderable subset. The A -sized copies just constructed are new.

All these segments have finite positive powers of cardinal A and new countable power of cardinal $\mathbb{R}_N$. They therefore yield no strict cardinal descent among themselves. A further recorded refinement says that each final segment has an old countable order-dense subset, using the common-divisor and disjoint-cone properties of the merger rays. That refinement has a written proof in [N9], but its separate independent audit is not part of the saved segment audit; it is not used in any theorem above.

5.8 Boundaries of the current classification

The unresolved global questions in N are PP, the strong quotient property $Q(A)$, NDS, FB, the assertion that every cardinal antichain is finite, and BPI. Also unresolved are $\mathbb{R}_N \preceq A$, KWP_1 , the Ordering Principle, and Choice for arbitrary families of finite sets. The original model's countable-member choice consequence has not been transferred to N . Adding a native linear extension does not establish the Order Extension Principle for every partial order, and neither it nor the local fields settles BPI.

Two bounded deductions sharpen the outstanding work. First, a hypothetical map $j : \mathbb{R}_N \rightarrow A$ has a support root d such that $j[R_a] \subseteq aT$ whenever $a \leq_T d$. This produces a coherence requirement among local μ -sized injections, rather than a cardinal contradiction. Second, every directed ideal of T in N has a canonically chosen countable cofinal sequence: select least points in the old T -wellorder with heights approaching the supremum, then take finite joins. The positive-cost inequality rules out an omitted divisor. Hence all these ideals inject into $\mu^\omega \approx \mathbb{R}_N$ [N10]. These are saved local deductions; neither supplies the missing global coherent selector.

The original quotient proof explicitly requires support descent and coherent matching across charts. New availability ideals need not be ground objects, and arbitrary ordered three-chart amalgams can cycle. A bound on local names or ideals by $\mathbb{R}_N$ is therefore not a proof of an injective presentation theorem. Likewise the Truss-model comparison in Ryan-Smith's work shows why $\text{AC}_{\text{WO}} + \text{SVC}(\mathbb{R})$ alone must not be treated as PP.³

³C. Ryan-Smith, "Local reflections of choice," *Acta Mathematica Hungarica* 176 (2025), 244–257, <https://doi.org/10.1007/s10474-025-01533-3>. The relevant comparison is Proposition 3.19 in the published version, numbered 3.18 in the author-hosted preprint. It concerns a different model; it is not a negative PP theorem about N .

Saved proof and status register for this section

All filenames below are in the public archive directory `evidence/original_model/`, mapped from the workspace record `outputs/BPI_RPP_five_hour_research/`. The source references are the pinned OpenAI construction in `work/openai_math/`: in particular `forcing.tex` (diagram amalgamation, local decision, ordinal choice and no countable ground sequences), `monoid.tex` (probe gates, merger normal forms and the construction schedule), and `quotients.tex` (availability ideals and coherent quotient matching). The external construction is credited to <https://github.com/openai/math>; the extensions and deductions reported here must be distinguished from statements made in that manuscript.

[N1] **Chain conditions and chart reduction, audited.** `FINITE_LINEAR_EXTENSION_FORCING_CCC.txt`; `FINITE_LINEAR_EXTENSION_CCC_INDEPENDENT_AUDIT.txt`; `FINITE_ORDER_FORCING_CCC3_CHART_REDUCTION.txt`; `FINITE_ORDER_FORCING_CCC3_INDEPENDENT_AUDIT.txt`.

[N2] **Preservation, audited.** `BOOLEAN_COMPLETION_COVERING_AND_HARTOGS_PRESERVATION_AUDIT.txt`; `CCC3_PRESERVES_AC_WO_CONDITIONAL_AUDIT.txt`; `AC_WO_DC_PRIMARY_SOURCE_RECONCILIATION.txt`. The “conditional” filename records an earlier checkpoint: its CCC_3 hypothesis was subsequently verified in [N1].

[N3] **The old complete Boolean algebra, audited.** `COMPLETE_CCC_BOOLEAN_ALGEBRA_EXACT_CARDINAL_A_AUDIT.txt`; `BOOLEAN_ALGEBRA_ROOT_AUDIT.txt`.

[N4] **Real maps and thresholds, audited.** `GENERIC_ORDER_PROFILES_INJECT_A_INTO_REALS.txt`; `GENERIC_REAL_PROFILE_INDEPENDENT_AUDIT_AND_QUOTIENT_TEST.txt`; `BOOLEAN_CODES_FOR_NEW_REALS_ROOT_DRAFT.txt`; `BOOLEAN_CODES_REAL_QUOTIENT_INDEPENDENT_AUDIT.txt`. The retained “draft” filename is accompanied by the separate audit.

[N5] **Local fields and localization, audited.** `LOCAL_REAL_FIELDS_REFLECT_NATIVE_ORDER.txt`; `LOCAL_REAL_FIELDS_INDEPENDENT_AUDIT.txt`; `LOCAL_ZFC_REAL_COVERS_AND_ORDINAL_LOCALIZATION_AUDIT.txt`.

[N6] **Old comparisons and relative finite bases.** `OLD_CARDINAL_COMPARISONS_AND_RELATIVE_FINITE_BASIS.txt`; `OLD_SURJECTIVE_COMPARISONS_AND_RELATIVE_PP.txt`. These are mathematical deductions from the audited preservation and normal-form results; the surjective comparison has a recorded second reconstruction.

[N7] **Nonsplitting and multisections, audited.** `COUNTABLE_MULTISECTIONS_AND_NONSPLITTING_PRESERVATION.txt`; `COUNTABLE_MULTISELECTOR_INDEPENDENT_AUDIT.txt`.

[N8] **Choice bounds and continuum-function failure.** `NEW_MODEL_KWP_OP_FINITE_CHOICE_BOUNDARY.txt`; `FINITE_KWP_ICF_COLLAPSE_INDEPENDENT_AUDIT.txt`.

[N9] **Resolved segment cardinalities, audited.** `ANCHORED_RAYS_RESOLVE_TAIL_AND_INTERVAL_CARDINALS.txt`; `GENERIC_ORDER_SEGMENTS_INDEPENDENT_AUDIT.txt`. The preliminary file `GENERIC_LINEAR_ORDER_SEGMENTS_AND_SQUARE_ROOTS.txt` now carries a supersession notice. Its open tail alternative is not a current unresolved question. The separability refinement in the later note remains separately flagged above.

[N10] **Bounded further deductions and remaining gaps.** `REAL_TO_LABEL_INJECTION_CHART_LOCALITY_AUDIT.txt`; `NEW_AVAILABILITY_IDEALS_REAL_CODE_BOUND.txt`; `TRUSS_QUOTIENT_COMPARISON_AFTER_ORDER_FORCING.txt`. These do not assert a completed new quotient classification.

6 Earlier permutation-model results and construction obstructions

These results belong to the earlier ordinary Fraenkel–Mostowski (FM) search. They are retained because a failed PP construction may still satisfy useful restricted principles, and because the obstructions explain why merely modifying a group or filter is difficult. The ground universe has atoms and choice. The action is faithful, the normal filter is effective, and “open” means allowed by that filter. All densities, indices and multiplicities in this section are external ground cardinals. Claims about specific constructions are conditional on their recorded normal-form and extension lemmas; they are not claims about every permutation model.

6.1 Exact and sufficient group criteria

The exact PP criterion. For every open K , there must be one open $J \leq K$ such that for every pair of open subgroups $L \leq H \leq K$ there is a J -equivariant injection

$$\text{Res}_J(K/H) \hookrightarrow \text{Res}_J(K/L). \quad (\text{P})$$

The same J must work for all pairs. Matching requires the exact conjugacy type of each point stabilizer and enough target orbits of that type. A fixed point, an inclusion between stabilizers, or individual repairs with unbounded supports do not suffice. The criterion is hereditary to open subgroups; passing to a dense subgroup with the induced topology preserves it.

Common compression. The group formulation of AC_{WO} used here is

$$\forall K \text{ open } \exists Q \leq K \text{ open } \forall L \leq K \text{ open } \exists k \in K \quad k^{-1}Qk \leq L. \quad (\text{C})$$

In a faithful effective action, failure of choice is equivalent to the identity subgroup not being open. A coset object K/H is internally well-orderable precisely when its normal core in K is open: a supported well-order has trivial permutation action on its domain, and an open core conversely supports a ground well-order.

Uniform non-well-orderable profiles. A sufficient additional condition is that, after one open refinement $J \leq K$, all K/H with non-open core become isomorphic as J -sets. Combined with (C), well-orderable targets are handled by simultaneous sections and the remaining targets by these common bijections. This implies (P). It also gives a two-prototype decomposition and hence NDS. No nondiscrete realization of this uniform condition was obtained.

Finite prototypes. More generally, suppose that for each open K there are an open $J \leq K$ and finitely many continuous J -sets $U_1, \dots, U_r$ such that every restricted K/H is a disjoint union of ground-cardinal multiples of these same prototypes. Decomposing a supported sequence into such orbits gives coefficient vectors in a finite product of well-orders. An increasing pair of vectors gives a supported reverse comparison, excluding strict descent. This proves NDS and finite antichains, without a uniform finite bound on their sizes. The condition also implies (C), by fixing a point of each nonempty prototype. Its realization remains unresolved.

Relative point completeness (RPC). For a transitive atom action with cofinal point fixers, (P) requires a refinement below which every open subgroup is the exact stabilizer of an atom. Equivalently, at a suitable root b , every open $H \leq G_b$ has the form $\text{Fix}(b, a)$. In particular such a subgroup is determined by its fixed atoms; arbitrary increasing unions must retain this property. In this setting, (C), RPC and the dichotomy that every internal atom subset is well-orderable or

equipotent to the atom set imply full PP. This sufficient criterion has no realized nondiscrete example in the record. A multi-HNN construction realizes (C) and RPC with failure of AC, but fails the dichotomy, PP and NDS.

6.2 Necessary topology and a sharp atom-model boundary

The following restrictions have saved proofs and separate checks.

1. Under (C), the intersection $R(K)$ of the open normal subgroups of K is open and satisfies $R(R(K)) = R(K)$. It has no proper open subgroup of finite index. A commensurated open subgroup has open core; a cofinal commensurated neighbourhood base therefore forces discreteness.
2. A nondiscrete (C)-group has no open compact or open Roelcke-precompact subgroup. Cofinally one obtains an open $B \rtimes \langle g \rangle$ with quotient $\mathbb{Z}$. The contraction argument gives a proper conjugate copy inside a common-compression witness; co-Hopfian, Jónsson, or malnormal witness designs are consequently unsuitable in the stated setting.
3. A nondiscrete effective FM group satisfying PP has

$$d(K) \geq 2^{\aleph_0} \quad \text{for every open } K.$$

The proof interpolates a rationally dense chain in one open-subgroup interval. Real cuts yield continuum many open intermediate subgroups, each requiring a distinct exact stabilizer in one coset target. A dense subset meets all open cosets, giving the cardinal bound. Thus no separable open subgroup is possible. This excludes Polish effective topologies, not every topology on a countable atom set.

4. If J is a common-compression witness in K , then $[K : J] = d(K)$ and the character and weight are bounded by this index (the weight equals $d(K)$ in the recorded setting). Below a PP witness, the number of open subgroups is at most $d(K)$. A finite-prototype witness satisfies the same local capacity bound. No general amplification to $2^{d(K)}$ was proved.

Finite choice in ordinary FM. For each fixed $n \geq 2$, $\text{AC}_{\text{WO}} + \text{AC}_n$ implies AC in the ordinary FM setting above. Here AC_n means choice for arbitrary families of n -element sets. The proof uses tail-equivalence classes of injective atom sequences. Ordinal-indexed choice supplies a common support for representatives of well-ordered configurations. If that support still moves an atom, a contracting translate creates an exact n -cycle of tail classes, contradicting a supported selector. Therefore the support fixes all atoms and is trivial.

Consequently $\text{AC}_{\text{WO}} + \text{OP}$ collapses to AC in ordinary FM. The OpenAI generator $A \subseteq \mathcal{P}(\kappa)$ has a lexicographic order; its full linearly orderable strong-generator package cannot occur in a nonchoice ordinary FM model. This is an atom-model obstruction, not a contradiction in pure ZF.

If an unordered strong generator did exist in ordinary FM, it could be replaced by a subset of atoms of least external nonchoice size λ . After an open restriction, its point stabilizers would form a cofinal neighbourhood base, and every further open subgroup would have density exactly λ . Thus $\lambda \geq 2^{\aleph_0}$. These constraints do not construct such a generator or prove that it is impossible.

6.3 Germ and monoid completions: what succeeded locally

In the imported free germ construction, complete stationary-tail quotients and countable-prefix quotients admit reverse injections into the atom set with one prescribed root support. The proof

preserves exact relative stabilizers through triangular extensions and provides enough tagged target orbits. This is a whole-quotient result, not just a local finite test.

Proper-suffix, nonfree whole-act extensions improve this to exact codes for ordered tuples and unordered finite configurations. In the mixed completion, each nontrivial supported atom orbit contains a copy of the whole atom set after one common refinement. Hence every atom subset is well-orderable or equipotent to A , and PP and NDS hold when restricted to atom subsets. With the recorded countable-code schedule, $\text{Count}(A) \cong A$ and every surjection from an atom subset with countable fibers has a reverse injection. The argument does not choose enumerations of all fibers simultaneously.

Zero-suffix markers repair the residual stabilizer groups of the proper-suffix construction: a fair schedule kills every final group $U(P_f)$, where $P_f = \{u : uf = f\}$. Basic root fixers then have no proper open normal subgroup. Arbitrary open subgroups may still have nontrivial quotients. Neither this repair nor countable-fiber PP establishes full PP.

6.4 Explicit counterexamples to the germ candidates

Free finite-period germs. A preserved divisibility invariant says that certain periodic fixed labels meeting a designated cone already lie in that cone. It leaves a target coset with an exact relative stabilizer absent from every possible image point. A stronger direct witness is

$$\text{Anc}(u) \times M \twoheadrightarrow A, \quad (x, r) \mapsto xr,$$

where ancestors of u have no supported reverse injection from A . Thus the final imported Section 85 candidate fails PP, including its optional local-group completion. Global absorber resets do not remove the endpoint-specific invariant. This proof does not establish failure of NDS in that candidate.

Proper-suffix-only completion. It repairs that ancestor witness but preserves an action-groupoid residual. Tagged loops yield 2^λ open overgroups of one point fixer, while the relevant target coset has only λ points. Exact stabilizer capacity refutes full PP. Its restricted positive results remain valid; NDS is unclassified here.

Mixed zero-suffix completion. The residual-group obstruction is repaired, but a long-chain quotient supplies a different counterexample. The scheduled operations produce decreasing principal ideals $h_\alpha M$ with

$$\bigcap_{\alpha < \omega_1} h_\alpha M = aM.$$

The relation xEy iff $xh_\alpha = yh_\alpha$ for some α defines an internal surjection $A \twoheadrightarrow A/E$. Its class stabilizer is a proper increasing union with the same fixed atoms as a larger root fixer. A supported reverse injection is impossible. The construction generalizes at the stated regular scheduling boundaries. Countably based repairs preserve this obstruction; arbitrary uncountably based repairs are not excluded. Failure of NDS does not follow from this PP counterexample.

6.5 Fields, local-motion groups, and symmetric iterations

Construction or method	Precise outcome recorded
Countable full-field group	Dense chains of root fields give continuum many distinct cut fixers in an interval whose source enumeration orbit is countable. PP fails; the literal uncountable-chain proof is unnecessary. NDS remains unclassified.
Uncountable algebraically closed atom field with countable closed-subfield supports	A proper dense subgroup of a countable field automorphism group has the same fixed atoms as its closure. The enumeration quotient violates exact stabilizer matching, so PP fails.
Saturated full separably closed fields	With the recorded named finite p -basis and sufficient saturation/homogeneity, every fixed finite or infinite support bound is excluded for PP. Kummer root capacity, dense restriction groups, or a binary-profile surjection handle the separate size cases. Arbitrary normal filters and restricted nonsaturated groups are outside this conclusion.
General saturated structures	Countable language/base with finite supports fails AC_{WO} under the stated saturation hypotheses. For language/base of size $\leq \lambda$ and all $\leq \lambda$ supports, order-indiscernible restriction groups violate PP capacity.
Increasing homeomorphisms of a ray	With initial-segment fixers, (C) and nonchoice hold and every open subgroup has continuum density. The endpoint quotient refutes PP; separate prime-localization coset tails refute NDS. First countability alone is not enough.
Finite piecewise-affine interval exchanges	Bounded-interval supports give (C) and nonchoice. Parameter subspaces give too many overgroups for PP; prime-cycle tails independently refute NDS. Removing an atom order does not repair the problem.
Continuous local-motion repairs	On the stated second-countable patches, a countable moving subgroup has the same fixed atoms as a larger patch group, giving the quotient obstruction. In the affine example, compact motion is dense, so a Hausdorff quotient killing it is trivial.
Full-symmetric enlargement	A genuine non-open, trace-preserving extension supplies exact old stabilizer types and repairs enlarged old quotient diagrams. The larger group creates new PP and NDS obstructions.
Iterated full-symmetric absorption	At cofinality above the support bound the limit is dense in a full symmetric group and fails PP and NDS. At smaller cofinality, a reservoir diagonal defeats (C) and gives a well-orderable-target PP obstruction. Small-cofinality NDS is not settled.

A separate algebraic lemma is useful in the field exclusions: if an uncountable field C lies in a separably closed field with a transcendental t over C , the latter requires at least $|C|$ field generators over C . Roots of $t - a$ for distinct $a \in C$, after the stated finite cyclotomic adjustment, give the Kummer degree bound. The conclusion concerns literal field generation, not a closure operation that already includes separable closure.

6.6 Coherent coding is possible, but the constructed models fail

The forest constructions realize actual equivariant coding maps, so their positive part should not be confused with a merely proposed coding scheme.

1. A binary forest realizes $[A]^2 \cong A$ equivariantly; fixed finite arity m similarly realizes $[A]^m \cong A$. Appropriate infinite support filters yield (C) and nonchoice. A fresh cyclic configuration of prime order greater than m refutes PP, and tails over distinct such primes refute NDS.
2. A general finite-carrier theorem removes any bound on signature size: if every atom has an actual finite pointwise support under an embedded full basis permutation group, the quotient of injective sequences by finite coordinate permutations is a PP counterexample under the stated infinite support bound. Its fibers are countable. A merely finite carrier label is not enough for this theorem.
3. A countably branching forest realizes codes for all countable subsets and exact sequence codes. It repairs the previous finite-reordering quotient and finite prime-cycle tails. Nevertheless the fixed-support construction has too many open overgroups for PP.
4. NDS in that countable-code model fails by a different mechanism: hidden index- p subgroups of compact auxiliary product groups have the same fixed atoms as the whole product. The exact coded finite covers give strictly descending tails over distinct primes. Higher fixed infinite arities and the singular-bound version admit the corresponding capacity argument.
5. At singular λ , the recorded injective $< \lambda$ codes bootstrap to exact λ -sequence codes, causing support collapse and the previous obstructions. At regular uncountable limit λ , the particular forest fails common compression. A general theorem excluding all unbounded-arity or huge-signature constructions was not proved.

Proof records. The supporting archive includes the original follow-up mathematical notes under `evidence/earlier_FM/`, including `NO_SEPARABLE_OPEN_GROUP_PP_THEOREM.txt`, `RELATIVE_POINT_COMPLETENESS_CRITERION.txt`, `MIXED_COMPLETION_LONG_CHAIN_PP_COUNT_EREXAMPLE.txt`, `SATURATED_SCF_FIXED_SUPPORT_BOUND_NO_GO.txt`, `COHERENT_COUNTABLE_CODES_AND_CAPACITY_OBSTRUCTION.txt` and `HIDDEN_PRIME_COVER_NDS_OBSTRUCTION.txt`. Earlier “current candidate” labels in those chronological notes are superseded by the conclusions above.

6.7 Other imported forcing results and their limits

The untouched imported report also records restricted pure-ZF forcing analyses. These were historical source material, not a second independently certified construction in the present consolidation. Their retained conclusions are:

- Ordinal-target PP is equivalent to AC_{WO} ; the first failure of well-orderability in pure $\text{ZF} + \text{AC}_{\text{WO}}$ occurs at a power set of an ordinal. Under $\text{SVC}^+(S)$, full PP localizes to AC_{WO} plus PP between subsets of S . Whole-source $\text{PP}(S)$ is a separate condition when only $\text{SVC}(S)$ is known.
- Repairing old real quotients, keeping old reals, or proving PP between old objects does not control new sources. Externally ccc repairs and internally well-orderable repairs have distinct exclusions over the specified Feferman seed.
- The recorded stabilized repair handles quotients of the full final real set, with dependent choice and $\text{SVC}^+(\mathbb{R})$ under its stated assumptions. Its unrestricted ordinal-

choice/distributivity and arbitrary-source obligations were not closed. A sparse variant fails PP and the relevant ordinal-choice principle; that does not refute the different unbiased version.

- Specified Sacks, Silver, Mathias, Laver, Miller and related routes are excluded only under the report’s seed and preservation hypotheses. Ordinary splitting and fat splitting are not completely classified there. A no-proper-intermediate theorem retaining the seed does not cover submodels that discard old reals.
- An amoeba proposal lacks its countable-pattern properness proof; the full-condition product proposal has an explicit choice cone whose density is not proved. These are unfinished constructions, not consistency theorems.

Bounded-rank Jech–Sochor transfer alone cannot turn an ordinary-FM statement with unrestricted all-set quantifiers into the desired pure-ZF theorem. The positive pure model used in the current results is the separately cited OpenAI construction.

7 Verification, novelty and the remaining research programme

7.1 What was checked formally

The saved isolated build compiled all 75 unchanged pinned `PartitionConsistency` proof files with Lean 4.34.1 and the pinned Mathlib. The recorded `#print axioms` checks cover the main consistency declarations and the stronger inputs: non-well-orderable quotient/image bijections, ordinal presentation, and ordinal-indexed choice. Only `propext`, `Classical.choice`, and `Quot.sound` occur in those checked declarations. The construction sources were identity-checked. This is a local build of the cited formalization, not a Comparator challenge-definition comparison.

The ordinary relative-consistency conclusion still uses the first-order interpretation in the manuscript’s Section 8. Ambient Lean choice is part of the metatheory; it must not be confused with choice holding inside the modeled universe. Nor does a successful build of the original construction formally verify every new corollary in this report.

Two small additional developments were checked: `SixCycleAudit.lean` formalizes the finite cyclic-overlap and copied-assignment lemmas used in the BPI proof; `FiniteHeightAudit.lean` formalizes a finite-height combinatorial part of the order-forcing chain-condition argument. Their scope files identify the boundaries. Neither file formalizes the full forcing theorem, the entire six-cycle argument, the preservation theorem, the NDS deduction or the new exponentiation theorem.

7.2 Mathematical and finite checks

Separate AI-assisted reviews checked the finite-basis proof, exact comparison and lattice arguments, monoid-action and nonsplitting arguments, the six-cycle proof and its support/Cohen compatibility, parameter variants, cardinal-function coding, and the stated finite-order-extension results. The countable-member choice consequence received a second independent AI reconstruction during this consolidation. These checks can expose gaps and mismatched quantifiers, but they are not independent human peer review.

Finite computational checks validate the small graph gadget and the stated finite-order compatibility patterns; the saved files include the exact scopes and counts. Such finite checks do not replace the infinite forcing argument. Compilation, reference checks and visual PDF inspection concern the document, not mathematical truth.

7.3 Novelty boundaries

The literature record is a bounded search dated 6–7 October 2026. It includes a keyword search of the released text-source corpus across 722 OpenAI manuscripts (9,015 retrieved text-source files, hash-checked), followed by review of candidate matches. This was not a semantic proof-by-proof reading of all manuscripts. The target PP construction was examined directly.

Claim or subject	Attribution boundary
PP with failure of AC	The OpenAI construction; not a new model constructed by this report.
CSB* and WPP	Already explicit in the cited source, Remark 7.3.
Finite basis, NDS, finite cardinal antichains in W	No earlier explicit application located in the bounded search. The normal-form mechanism is extracted from the source, and the ordinal-vector argument is classical. No priority certificate.
BPI/OEP failure in W	No explicit statement located in the checked target source or bounded follow-up search. The new argument is a candidate additional application, pending specialist assessment.
Cardinal lattice and additive absorption	Related weaker consistency results are classical; broad novelty must not be inferred from a fresh proof or new notation.
Comparable power-set cardinals without AC	Already obtained in the first Cohen model in the cited Glazer discussion; not new as a bare consistency result.
Unequal exponents with equal powers at every base	Older unequal mutually surjective examples provide this phenomenon without PP. The present argument uses a different mechanism, with a well-orderable exponent and PP. Its exact prior status remains subject to further checking.
Real Hartogs/ordinal-choice package	Earlier comparison models exist. The new finite-order forcing analysis is the particular object studied here; no first-ever claim for its broad choice/real-cardinal package.
RPP versus BPI	Classical comparisons precede this report; merely obtaining the ordinary RPP here is not a new general implication.

The distinction between injection and surjection, members and indices, set families and sequences, and ZF and ZFA is essential when searching for prior results. A statement under a different formulation may require a choice principle to transfer. The source-conflict records retain cases where a later introduction appeared to contradict older published results; no new theorem is based on the disputed historical claim. In particular, this report does not advertise consistency of $DC + BPI + \neg AC$ as newly obtained.

7.4 Open questions retained at the cut-off

1. Does PP imply NDS or FB in arbitrary ZF models? Does NDS or FB imply PP? The model-specific positive package settles none of these unrestricted directions.
2. Is there a different pure model of $PP + BPI + \neg AC$? Does $SVC(A) + Q(A) + AC_{WO} + BPI$ force choice? The negative BPI theorem for W alone does not answer either question.
3. In $N = W[G]$, does $\mathbb{R} \hookrightarrow A$ hold? Are all non-well-orderable images of A equinumerous with A ? Do PP, NDS, FB or BPI hold? Preserve the distinction between the old-set comparisons already proved and these global new-set questions.

4. Can the new real-sized bound on directed availability ideals replace the original ground-cardinal bound in the coherent quotient argument? A coding bound alone does not give the needed supported matching.
5. What are the exact two-colour triple Ramsey thresholds? The finite-colour spectrum and the failure of fixed coarsenings to supply the desired binary obstruction do not settle this.
6. Can an unordered ordinary-FM generator realize the exact coset criterion, RPC plus the missing dichotomy, or the finite-prototype criterion? The many exclusions are architecture-specific and do not establish a universal impossibility theorem.
7. Which exact all-base exponent-cancellation or exponent-separation principles are equivalent to AC or to known weak-choice principles? The present model refutes the stated principles, and the recorded implications are limited to those proved above.

7.5 Files, authorship and publication record

The public package supplies this PDF, its complete LaTeX source, supporting mathematical notes, the two auxiliary Lean developments with check records, and a machine-readable source map with checksums. The evidence is grouped by original-model results, earlier FM investigations, source verification and novelty records. Chronological notes may retain an earlier unresolved status that was later settled; this report is the status guide, and the filenames cited in each chapter lead to the detailed arguments.

The original imported PDF and TeX remain unchanged in the local workspace. They are not included in the public evidence archive, which contains the mathematical follow-up records rather than the raw imported conversation or account/usage data. OpenAI's third-party manuscript and Lean source distribution are linked by pinned revision, not relicensed or republished here.

The earlier focused finite-basis paper remains available; this report extends the scope rather than overwriting its historical public version. The Weak Choice Atlas website is a separate software/catalogue output. Its search, comparison and provenance checks do not certify every mathematical result, and publication of this report does not imply that all its new records have already been added to that website.

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